

Sponsor Reputation and Capital Structure Dynamics in Leveraged Buyouts

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Received: March 31, 2023

Revised: February 1, 2024; April 29, 2024

Accepted: May 3, 2024

Published Online in Articles in Advance:
October 17, 2024

<https://doi.org/10.1287/mnsc.2023.00971>

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Abstract. We examine whether leveraged buyout (LBO) sponsors' reputations as borrowers affect the refinancing terms of their portfolio companies. In 510 U.S. LBOs for which we can reconstruct debt financing activity, 67% of financing events occur one quarter before the earliest existing debt maturity. These *On Time* events generally improve borrowing terms, whereas *Early* events feature more dividends, leverage, and higher cost. In each case, dividend issuance decreases and cost increases with the proportion of the sponsors' recent exits that are failures. Effects are absent for matched firms and prior to the LBO and are procyclical; sponsors with recent failures miss opportunities to decrease financing costs in good times.

History: Accepted by Bo Becker, finance.

Supplemental Material: The data files are available at <https://doi.org/10.1287/mnsc.2023.00971>.

Keywords: leveraged buyout • borrower reputation • private equity

1. Introduction

Our extensive knowledge of the dynamics of the capital structure of public firms¹ may not always apply to private firms, and, in particular, to the targets of leveraged buyouts (LBOs). In an LBO, a sponsor, often a private equity firm specializing in such deals, buys a company and finances a large part of the purchase price with debt backed by the target company's assets. Axelson et al. (2009) show that raising a buyout fund to make equity investments but financing each portfolio company's debt separately minimizes agency conflicts between fund managers and investors. In practice, the sponsor is an intermediary between lenders and the company and cultivates a reputation by owning many companies over multiple funds, thus creating a series of investment opportunities for lenders.

LBO sponsors perform exhaustive due diligence before bidding on their targets. Given the publicly observable nature of defaults and the strong incentives embedded in sponsors' compensation structure, lenders may believe that monitoring the sponsor partially substitutes for monitoring the borrower, with the absence of past default being the most positive signal (Diamond 1989). In their model of LBO capital structure, Malenko and Malenko (2015) distinguish between sponsors' skill for operational improvement and their reputation for refraining from diverting value from creditors in bad times. They predict that operationally skilled sponsors can borrow more because the value of

preserving their future LBO dealmaking exceeds the value of current opportunistic behavior.

In practice, operationally skilled sponsors might be more difficult adversaries in times of distress. Examples of protracted legal battles between creditors and sponsors include the buyouts of Aleris,² Caesar's Palace,³ iHeartMedia,⁴ J. Crew,⁵ Nine West,⁶ Neiman Marcus,⁷ and PetSmart.⁸ Thus, it is unclear how LBO lenders consider signals of distress in one portfolio company in their lending decisions to other companies that the sponsor manages. Moreover, a sponsor's reputation for protecting lenders' investments may evolve due to human capital turnover and growth in successive fund sizes: Ivashina and Lerner (2019) find that 9% of original senior partners and 12% of junior partners have left the firm by the time the next fund is raised, and Harris et al. (2023) find that there no longer is persistence in successive LBO fund returns.

Most work on LBOs, summarized in Table 1, focuses exclusively on their initial capital structure and is not ideally suited to answer these questions. The few exceptions⁹ do not examine the dynamics, composition, or the cost of post-LBO debt. This gap in the literature is puzzling in light of the finding of Roberts (2015) that a typical public firm's bank debt is renegotiated every nine months and because of the particular importance of debt in the capital structure of leveraged buyouts. It is plausible that many LBO targets seek to refinance as conditions change, either in order to continue to afford the payments if the target underperforms, to preserve

Table 1. Summary Statistics from Other Empirical Work on LBOs

	Number of LBOs	LBO dates	Location	Public to private, %	Inception debt/EV	Postbuyout debt/EV	LBO data	Debt data
Kaplan (1991)	76	1980–2006	United States	100	0.85	0.978 ^a	SDC	Compustat
Demiroglu and James (2010)	180	1997–2007	United States	100	0.6–0.7 ^b		SDC	Dealscan
Guo et al. (2011)	192	1990–2006	United States	100	0.7		SDC, Dealogic	SEC filings
Ivashina and Kovner (2011)	1,590 ^c	1996–2005	United States	100			Dealscan	Dealscan
Boucly et al. (2011)	839	1994–2004	France		0.58 ^d		SDC, Capital IQ	Benefices Reels, Normaux
De Maeseneire and Brinkhuis (2012)	126	2000–2007	Europe		0.65		A bank	A bank, Dealscan
Axelson et al. (2013)	1,157	1980–2008	World	32.1	0.67		Capital IQ	Dealscan, Amadeus
Cohn et al. (2014)	317	1995–2007	United States	100	0.433	0.451	SDC	IRS
Ayash and Schutt (2016)	183	1980–2006	United States	100	0.31	0.68	SDC	Edgar
Achleitner et al. (2018)	130	2000–2008	Germany	1.5	0.61–0.70 ^e		Three banks	Three banks
Haque et al. (2022)	4,546	2013–2020	United States		0.492	0.544	Preqin	Y-14
Cohn et al. (2022)	288	1995–2009	United States		0.575	0.7–0.9 ^f	SDC, Capital IQ	IRS
This paper	510	1988–2021	United States	29.2	0.636	0.792	Preqin	Dealscan

Notes. This table provides summary statistics on selected empirical studies on LBO capital structure, ordered by publication date. Postbuyout debt/EV reports any measure of postbuyout leverage regardless of the year in which it is measured. In the data columns, SDC, Securities Data Company database. Dealscan is Loan Pricing Corporation's Dealscan database. Amadeus signifies Bureau van Dijk's Amadeus database.

^aDebt to book value for 33 deals in fiscal 1989.

^bTable 2, p. 313.

^cNumber of loans.

^dTwenty-two deals.

^eFrom Figure 2(a).

^fMedian in Years 1–3 after buyout (Figure 3).

managers' incentives if the target outperforms, or to finance mergers and acquisitions (M&A) activity.

Our paper makes several contributions to the empirical literature on corporate capital structure and to the LBO literature in particular. The first is descriptive. Among the 510 buyouts for which we can reconstruct the history of financing activity until the sponsor's exit or until the third quarter of 2022, there are on average 1.582 financing events after the initial deal package. We find that 67% of financing events occur *On Time*, in the quarter prior to the maturity of the company's earliest maturing loan. One might expect that deal terms ease over time, conditional on the survival of the borrower, but perhaps the most successful LBO deals exit early. We find that follow-on debt financing deals generally improve borrowing terms relative to the initial package. *On Time* deals are associated with a 14.4% average increase in debt and a 11.1-basis-point average decrease in borrowing cost. For an average LBO company in our sample with an enterprise value of \$1,249M and a debt-to-EV ratio of 0.79 during the life of the deal, an 11.1-basis-point decrease in annual borrowing cost is associated with savings of \$1.078M per year, adding approximately $11.1 \times 0.79/0.21 = 0.42\%$ annual return to equity investors.

In contrast with *On Time* deals, *Early* deals are associated with a 37.9% average increase in debt and a 12.5-basis-point average increase in borrowing cost, and 25.4% of *Early* deals involve a dividend, which can add

significantly to equity returns, whereas only 8.62% of *On Time* deals involve a dividend. Thus, debt financing activity after the initial LBO package, even when its timing is forced by the maturity of existing debt, is strongly relevant to LBO fund returns.

When comparing a subset of LBO firms to matched, independent private firms and to their own pre-LBO activity, we find that financing events, including the mix between *On Time* and *Early*, are not significantly more frequent for LBO firms, although dividends are more than twice as frequent. Compared with matched firms, LBO capital structures throughout the deal life contain significantly more debt with higher spreads, longer maturity, and a lower proportion of revolving debt.

The second and main contribution of this paper is to test whether the sponsor's evolving reputation for protecting lenders' investments, separately from skill at generating fund returns and from the strength of the lender-borrower relationship, is rewarded by cost savings from intermediate financing activity. In 180 public-to-private LBOs from 1997 to 2007, Demiroglu and James (2010) find that LBO sponsors with longer histories, more past deals, and greater market share obtain better initial LBO credit conditions, and Achleitner et al. (2018) find that such sponsors obtain looser loan covenants. However, these results might be due to relationship building by banks in hopes of future fee business, as we

know of no evidence that large sponsors better protect lenders' investments.¹⁰ Moreover, the prior focus on the cross-section of initial LBO debt terms, coupled with near-static measures of reputation, leaves open the possibility of selection bias in which portfolio companies these sponsors acquire.

Our measure of (lower) skill at avoiding lender losses, *SponsorFailure*, is the proportion of the sponsor's exited deals over the last two years that have resulted in default or distressed recapitalization.¹¹ We do not distinguish between sponsors' propensity to divert or inject value once distress occurs (Malenko and Malenko 2015) and their propensity to incur downside risk in their portfolio companies, if the latter is not symmetric to upside risk as measured by *SponsorIPO*, the proportion of Initial public offerings (IPOs) among the sponsor's recent exits. Both diversion and downside risk can result in losses for lenders. Consistent with the findings of Braun et al. (2017) that bottom quartile deals tend to recur for a given sponsor, we find that *SponsorFailure* is positively related to its lagged value over nonoverlapping quarters, even controlling for measures of economic conditions.

We do not observe whether the borrower or lender initiates each financing event, and the decision to refinance in a given quarter may be endogenous to the outcomes that we observe. However, deals that occur in *On Time* quarters are likely to be driven primarily by the desire from both parties to refinance maturing debt.¹² The richness of our data and the length of our quarterly time series also allow us to control for several types of lending relationships¹³ that may affect financing deal terms. We separately measure the proportions of lenders in the syndicate that have a prior lending relationship with the LBO target company and with the sponsor. Ivashina and Kovner (2011) find that lenders that participated in a greater proportion of a sponsor's past loans provide better initial LBO loan terms, especially when that sponsor is likely to generate incremental banking revenues. Although we find evidence of the effect documented in Ivashina and Kovner (2011) in our subsample of initial packages of public-to-private deals, we do not find a strong broader role for the measures of relationship in subsequent financing events. This could be because we use a different measure¹⁴ or because that paper uses a much earlier sample period. A third variable measures the proportion of related lenders who are affiliates of the sponsors to the deal (Fang et al. 2013).

We find little effect of our measure of recent failures on the cross section of initial debt terms. This is perhaps unsurprising given that the predictability of sponsor failures is only significant over two years, whereas most buyouts have longer horizons. Examining the incidence of new financing activity within LBO deals, we find that renegotiations are less likely for infrequent

sponsors. This contrasts with the sample of public firms of Roberts (2015), where borrowers with more uncertain future prospects renegotiate more often. We also find that a one-standard-deviation increase in *SponsorFailure* is associated with a 19.2% lower probability of issuing a dividend in a given quarter. This complements the findings of Harford and Kolasinski (2014) that dividends are not typically associated with higher future incidence of distressed restructurings among LBO targets and suggests that the riskiest sponsors are denied the opportunity to issue a dividend.

Next, we examine the incidence of loosening (defined as an improvement in either cost, loan amount, or maturity, whereas other variables remain at worst unchanged) and tightening financing arrangements, conditional on a financing event. Tightening deals are only 6.7% of all financing deals, whereas loosening deals are 49% of all deals. A one-standard-deviation increase in *SponsorFailure* is associated with a 6% lower incidence of a loosening financing deal. Thus, LBO companies generally experience an amelioration of terms at these intermediate financing events and that sponsors' recent performance affects these terms.

To determine the drivers of the relation between *SponsorFailure* and deal terms, we examine loan amount and loan cost in turn. We find that a one-standard-deviation increase in *SponsorFailure* is associated with a two-percentage-point lower increase in debt at the time of the refinancing. This effect is driven by *Early* deals when banks' standards are, on average, loosening more compared with the last financing date. In those situations, a one-standard-deviation increase in *SponsorFailure* is associated with a 5.13% smaller increase in the loan amount, compared with a 37.9% average increase in debt for early deals.

Next, we examine the cost of newly issued debt. A one-standard-deviation increase in *SponsorFailure* is associated with an 8.5-basis-point increase in the weighted average loan spread, driven mainly by a 12.3 basis point increase in term loan spreads. For comparison, the average change in term loan spread is −13.3 basis points for *OnTime* financings and +2.98 basis points for *Early* financings. In comparison, Ivashina and Kovner (2011) find an eight-basis-point effect of their measure of repeat banking relationship strength. Our results remain when we compute *SponsorFailure* in alternative ways, retain only one intermediate financing event per LBO, and exclude deals that end in failure from our sample.

It is natural to ask whether lenders update correctly. A back-of-the-envelope calculation suggests that they do. An 8.5-basis-point increase in the cost of debt amounts to 50 basis points of increased revenue to the lender over the following six years, the average life of a loan. Because a one-standard-deviation increase in sponsor failure over the last eight quarters is associated with a 0.014 increase in *SponsorFailure* over the following eight

quarters, a loss-given-failure of $0.005/0.014 = 36\%$ would reconcile the increased revenue to the expected incremental loss for a risk-neutral lender. Although average values seem plausible, we also examine this question statistically. *SponsorFailure* at LBO inception does not predict whether the deal fails, echoing our prior finding that *SponsorFailure* is predictive over the following two years, but not longer. However, at the time of an intermediate financing event, *SponsorFailure* is predictive of whether the current deal ultimately fails. When we further control for the realized changes in lending terms at the time of the financing events, they subsume the predictability of *SponsorFailure* for the ultimate failure of the deal. This suggests that, on average, lenders accurately adjust for changes in the sponsor's probability of default by adjusting loan terms when LBO companies refinance their debt.

Lastly, we ask whether sponsor reputation is related to their ability to pick LBO targets or to their skill at managing companies. Using the finding by Kaplan and Schoar (2005) that fundraising is especially high for LBOs when the difference between the earnings yield on the S&P 500 and the high yield rate is large, we use this as a measure of competition for LBO targets. During competitive times, when multiple sponsors may be bidding on each target, the sponsor with the highest projected operational improvements may be more likely to win the deal. If lenders believe that sponsors' managerial ability is key to preserving their capital, one might expect the effects of *SponsorFailure* to be concentrated in LBOs initiated during *Competitive* times. We find that the results are of similar magnitude and significance for both *Slow* and *Competitive* times and conclude that lenders seek a combination of sponsors' picking and managerial ability or that these skills are highly correlated among sponsors.

These findings are economically important. If banks monitor the sponsor's evolving reputation for creditworthiness, dividend policy and refinancing terms for maturing loans of entire private equity fund portfolios may be affected by a recent distressed exit. This may increase correlations among the assets a sponsor manages and further reduce the incentives to make bad investments described in Axelson et al. (2009) when they show that sponsors are optimally incentivized by a private equity fund structure. The implications of these findings are severalfold. Economic models of LBOs should consider that intermediate financing activity adds nonnegligible value for equity holders. Further, in assessing optimal leverage levels for a given LBO and responses to distressed situations, a sponsor should consider the effects of a potential default on the near-term financing opportunities of other companies across their funds' portfolios. Lastly, this effect may decrease the diversification benefits of holding multiple companies in a fund.

2. Data and Variable Construction

2.1. Data

To build a sample of deals for which we will reconstruct the capital structure over time, we retain LBO deals from Preqin with a known size of at least \$100M and with a known sponsor ownership stake. We remove firms that are merged into others in less than two quarters and firms that are in finance, insurance, or real estate. We remove deals in which the sponsor assumes any of the target's existing debt, per Capital IQ.¹⁵

We require each deal to have loan data from the Loan Pricing Corporation's Dealscan database of syndicated loans, from which we hand-collect Borrower IDs by name. Some deals use an acquisition vehicle for the initial LBO with a different name, and we use Web searches to identify these when possible. Because Dealscan only covers syndicated loans that are used by larger companies, our results may not extend to smaller LBO targets. Table 1 shows that several related papers also use Dealscan. Ivanov et al. (2021) and Haque et al. (2022) study private firms using the Federal Reserve's Y-14 annual loan data, which has more coverage of smaller firms and includes balance sheet information. Annual data are not as well suited to our purposes because the quarter prior to maturity is a crucial refinancing quarter. Moreover, the Y-14 program was implemented in 2011 with data beginning in 2012, and a long time series is needed in order to follow many buyouts from inception to exit and to examine reputation effects.

Dealscan data are organized by loan packages containing individual loan facilities. The facilities in a package do not necessarily all start at the same time but were all agreed on at the same date. Letters of credit and other transactions that do not represent actual loans are removed. Refinancings of debt are often indicated by Dealscan as of the 2021 reorganization of the database, but some refinancings are left out, judging by the comment fields at the Dealscan package level, and we account for these by hand. We confirm that each refinancing is not associated with a new equity owner using the exit dates and by reviewing the comment fields in Dealscan. Online Appendix A provides more detail on the process of reconstructing the time series of the capital structure. We are able to reconstruct this time series for 510 LBO deals with inception dates between 1988 and 2021.

To obtain bond financing information, we hand-match the name of the LBO target to the list of issuers in Mergent. Mergent allows us to reconstruct the number of bonds outstanding at any given time for an issuer. We also hand-match by target name to the SDC database. Often, the name that SDC uses for the target is the one that was in use prior to a name change by the acquirer, while in Preqin the name is the newest name available. SDC sometimes refers to the target as a

subsidiary of the seller. Thus, when a simple match is not available, we search all deals that were announced on that date for deals with the same acquirer and similar deal amount, and we investigate whether there was a name change via a news search.¹⁶ We also hand-match target names to Capital IQ, which provides some annual financial statement data for private firms, mainly those with publicly traded bonds outstanding.

Other variables serve as controls, and their definitions appear in Table 2. To measure time-varying bank lending terms, we include $P_tightening$, a variable used by Axelson et al. (2013). This is constructed using the net responses to a quarterly Federal Reserve survey of senior loan officers of 80 U.S. banks and the U.S. branches of 24 non-U.S. banks, entitled *Net Percentage of Domestic Banks Tightening Standards for Commercial and Industrial Loans to Large and Middle-Market Firms* (Federal Reserve variable *DRTSCILM*). For quarters after the LBO's inception, we compute $\Delta P_tightening$, the difference between this value in the current quarter and the value at the time of the last financing event or the LBO's inception. Table 3 shows that $P_tightening$ is negative on average in our sample period, reflecting the fact that there are more quarters during our sample period when banks are loosening lending standards. For robustness, we use high-yield rates and the difference between the current high-yield rate and the rate at the time of the latest financing event instead of $P_tightening$ and $\Delta P_tightening$, and find similar results.

2.2. Measures of Sponsor Reputation, Skill, and Lending Relationships

2.2.1. Reputation and Skill. To measure sponsor reputation, we compute *SponsorFailure*, the proportion of the sponsor's deal exits that have ended in bankruptcy or reorganization in the last two years (quarters $q - 1$ through $q - 8$, for each quarter q) or since the LBO target's last financing event if this is shorter. This measure is computed using the Preqin Exits data, excluding minority investments, growth capital, private investments in public equity (PIPEs), mergers, corporate investments, and add-ons. This results in 10,510 deals from 1969 to 2021 that are plausibly leveraged and that can be attributed to one or more sponsors and with nonmissing exit dates and exit types. We choose a window of two years because for the median LBO-quarter, the sponsor has had 3.5 exits in that time frame, whereas in the last one year, there have been a median of only two exits, but Table B1 in the Online Appendix shows that the main results of the paper are robust to the choice of time frame.

Our sample of exits consists of 11,233 U.S. deals in the Preqin exits data, excluding minority investments, growth capital, PIPEs, mergers, corporate investments, and add-ons. We look up additional exit dates by hand using news searches and 10,510 deals from 1969 to 2021

in our sample of plausibly leveraged deals that can be attributed to one or more sponsors and with nonmissing exit dates and exit types.

We control for operational skill by including *Sponsor-IPO*, which is the proportion of exits that were IPOs in the last eight quarters or since the target's last financing. In the absence of infrequently available company-level returns, IPOs are a popular measure of success for private capital investments (Gompers 1996, Gompers et al. 2016, Howell 2017). Jenkinson and Sousa (2015) find that sponsors use early exits via IPO to advertise their skill. This is also a useful control variable because persistence in sponsor failures could be driven by persistence in the riskiness of the sponsors' deals, which could be captured by this measure of the right tail.

In order to distinguish the absence of failures or successes from the absence of exits, we also compute *SponsorNoHist*, an indicator for an absence of sponsor exits over the same period. *SponsorNoHist* is related to the measures of sponsor reputation in Demiroglu and James (2010), in particular, the frequency of past deals. In this paper, however, we are more interested in a measure of a lender's dynamic assessment of the risk that a deal will result in loan default. Although *SponsorNoHist* and *SponsorFailure* are linked in that frequent failures may affect the number of future deals that the sponsor can undertake, that process may be noisy and long. To capture and control for industry-level variation in LBO defaults, we construct an analogous variable *IndFailure* by scaling the number of failures by the number of exits over the last two years in each quarter for the 52 primary industries that Preqin identifies.

2.2.2. Relationships. We measure three aspects of relationship lending. *RelationshipLenderC* is the proportion of the current lenders that participated in at least one of the prior two loan packages for the target *company*. *RelationshipLenderS* is the proportion of the current lenders that participated in at least one of the prior two loan packages for the target *sponsor*. In contrast, Ivashina and Kovner (2011) measure relationship strength as the percentage of the amount of the initial LBO loans to deals that were underwritten by the lead bank in the last five years. Their sample comprises only public-to-private deals from 1993 to 2005, and their measure would be difficult to compute for our sample, for which complete Dealscan data are not always available throughout the LBO deal's life.

We also compute an indicator for whether the lender is the same entity as the sponsor, *RelatedLender*. Examples of occasional related lenders in our data are Apollo, Thoma Bravo, Jeffries, Goldman Sachs, Citibank, and KKR.¹⁷ *RelatedLenders* are typically not lead arrangers of the debt packages that they participate in. Nonbanks do not as of this writing have the ability to

Table 2. Variable Definitions

Variable	Description	Source
Δ	Quarter-to-quarter change in a variable. % Δ indicates percentage change.	
<i>Assets</i>	Total assets, available annually for a subset of deals, principally for portfolio companies that issue public financial statements. When computing % Δ Assets and Δ Debt_TA, four quarters lagged Assets to must be available.	Capital IQ
<i>Banks Looser, Banks Tighter</i>	A measure of P_tightening in the current quarter minus P_tightening in the quarter of the prior financing event, which could be the LBO's inception.	
<i>BankSponsor</i>	Indicator variable for whether an investment bank is sponsoring the deal.	Preqin
<i>BorrowingCost</i>	The weighted average borrowing cost of the current package and any outstanding bonds, updated quarterly based on repayment and current base rates.	Dealscan and Federal Reserve
<i>ClubDeal</i>	Target that is purchased by several private equity funds as a group.	Preqin
<i>DealValue</i>	LBO deal value in millions of dollars.	Preqin
<i>DealLength</i>	Deal length from the date of the first loan until the exit date in years.	Dealscan and Preqin
<i>Dividend</i>	Indicator for whether a financing event is associated with a dividend. The primary or secondary purpose of at least one loan during a deal-quarter must be "Dividend Recapitalization" or "Dividend" or Distribution to Shareholders."	Dealscan
<i>Debt_TA</i>	Total debt scaled by total assets. Total assets is updated over time.	Dealscan, Capital IQ
<i>Debt_EV</i>	Total debt scaled by initial EV. Initial EV remains the same over time.	
<i>Early</i>	A financing event occurs before one quarter prior to the maturity of the earliest maturing current loan.	Preqin
<i>EV</i>	An estimate of enterprise value at the time of the LBO: deal size scaled by the proportion acquired by the LBO sponsor.	Preqin
<i>Industry_DA</i>	The median debt to asset ratio in a given quarter for public firms in the same Fama-French 49 industry.	Compustat
<i>IndustryFailure</i>	The count of the number of bankruptcies of other portfolio companies in the same Preqin industry in the past two years or since the last financing deal if that is shorter, divided by the total number of exits in that time frame.	Preqin
<i>Maturity</i>	The weighted average number of years until maturity at loan initiation. Loan facilities are weighted by dollar amount outstanding. T indicates maturity for term loans, and R indicates maturity for Revolving credit. When several relevant loans exist, they are weighted by loan size.	Dealscan
<i>MnA</i>	An indicator for whether the financing event is associated with a merger or acquisition. Specifically, the primary or secondary purpose of at least one loan during the quarter must be "LBO", "MBO", "SBO", "Takeover", "Acquis. line", "Merger", "Defensive bid", or "Spinoff."	Dealscan
<i>N_</i>	The number of instances during the life of the LBO. For example, <i>N_</i> Dividend is the number of dividends during the LBO.	
<i>P_tightening</i>	Average among banks of the banks who are tightening loan standards in that quarter for large borrowers.	Federal Reserve (T10Y3M)
<i>OnTime</i>	A financing event occurs in the quarter prior to the maturity of the earliest maturing current loan.	Preqin
<i>Opaque</i>	The firm does not have data on Total Assets available in Capital IQ in the current year or quarter.	Capital IQ
<i>PublicToPrivate</i>	Buyout of a publicly traded firm.	Preqin
<i>Revolver</i>	Percentage of the firm's debt that is revolving debt. Available only for firms with reconstructed capital structures.	Dealscan
<i>RelatedLender</i>	Proportion of lenders in the current package who are affiliates of a deal sponsor. For example, if a deal is sponsored by Blackstone and a Blackstone credit fund provides some or all of the debt.	Preqin
<i>RelationshipLenderC</i>	The proportion of lenders in the current package that have participated in one or both of the two prior loan packages from this company (not necessarily with this sponsor as owner).	Preqin
<i>RelationshipLenderS</i>	The proportion of lenders in the current package that have participated in one or both of the two prior LBO loans from this sponsor.	Preqin

Table 2. (Continued)

Variable	Description	Source
<i>SponsorFailure</i>	The count of the number of bankruptcies of other portfolio companies of the deal sponsor in the past two years or since the last financing deal if that is shorter, divided by the total number of exits in that time frame, using the original sample of 11,233 deals. Averaged over lead sponsors for club deals. Lead sponsors are identified via news searches and Capital IQ.	Prequin, Capital IQ.
<i>SponsorIPO</i>	The count of the number of IPOs among the sponsors other portfolio companies in the past two years or since the last financing deal if that is shorter, divided by the total number of exits in that time frame, using the original sample of 11,233 deals. Averaged over lead sponsors for club deals. Lead sponsors are identified via news searches and Capital IQ.	Prequin, Capital IQ.
<i>SponsorNoHist</i>	An indicator variable for the absence of exits for a given sponsor in the past two years or since the last financing deal if that is shorter. Averaged over lead sponsors for club deals. Lead sponsors are identified via news searches and Capital IQ.	Prequin, Capital IQ.
<i>Spread</i>	The all-in-drawn spread in percent. T indicates spread for term loans, and R indicates the spread for Revolving credit only. When several such loans exist, they are weighted by loan size.	Dealscan

syndicate financing but can lend through credit funds that they manage.

3. LBO Debt at Inception and Subsequent Changes

If lenders focus on sponsors' reputation for not expropriating lenders (Malenko and Malenko 2015) or simply downside risk taken by the sponsor, we expect that they will penalize instances of *SponsorFailure*. This relation will be absent if they instead favor banking relationships in order to drive other business with the bank. If lenders care about a sponsors' operational skill or upside risk, we expect that *SponsorIPO* will be related to refinancing terms. Alternatively, none of these measures will matter if banks only consider each individual portfolio company's financial health.

3.1. At Inception and During the Sponsor's Holding Period

Table 3 presents summary statistics for the sample of 510 LBOs for which we can reconstruct the capital structure over time. Deal-level variables and initial values at the closing of the deal are summarized in columns (1)–(4), and data for subsequent quarters are summarized in columns (5)–(8). The average deal value, as reported by Prequin, is \$1.207B. Enterprise value (EV) is the deal value from Prequin scaled by the proportion of the firm that is acquired and is \$1.249B. If this proportion is missing in Prequin, we use the value given in Capital IQ. The similarity of deal value and EV reflects that most LBO deals in our sample involve 100% acquisitions, although we only require a controlling stake. Total assets average \$2.640B in the subsample of deals for which the data are available from Capital IQ.

Kaplan (1991) finds a median LBO holding period of 6.82 years. We find a shorter period of 4.582 years, but longer deals' capital structures are more difficult to reconstruct and thus are less likely to be in our sample, and 98 of our 510 deals are not yet liquidated as of the end of our sample period. Public-to-private deals comprise 29.2% of the sample, and club deals, in which there is more than one sponsor, comprise 31.0% of our sample.

Fang et al. (2013) find that 30% of their deals were bank-affiliated in the years 1983–2009, and they point out that the Volker rule imposed after 2009 drastically reduced the number of such deals.¹⁸ We use the list of investment bank sponsors from Fang et al. (2013, table 1) augmented with PNC Bank and accounting for variations in names.¹⁹ In our sample, 302 of the 510 deals were announced in 2009 or later, and we find that an investment bank controls one or more of the equity sponsors in only 4.9% of deals.

Table 3 shows that initial debt to EV averages 0.636, and in the subset of firms for which total assets are available, initial debt to total assets averages 0.562. These are fairly similar to the quantities from past work, which are tabulated in Table 1.

A typical initial LBO deal package in our sample consists of a revolving credit facility and one or more Term Loan B facilities. "A" term loans, which are typically amortizing loans that have a shorter maturity, are less common, as are bridge loans and other loans. The mean of the proportion of revolving credit, *Revolver*, is 18.2%. We see only the total size of the credit facility, but in practice, it may not be fully utilized. Thus, our estimates of total debt may be overstated relative to a data set that accounts for actual revolving debt utilization. The weighted average initial maturity of the

Table 3. Summary Statistics for Deal Level Variables and Quarterly Variables

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Quarter 0/deal level				Quarters 1–exit			
Variables	Mean	p50	Standard deviation	N	Mean	p50	Standard deviation	N
<i>Deal_Value</i>	1,207	650	1,924	510				
<i>EV</i>	1,249	681.1	1,994	510				
<i>DealLength</i>	4.582	4.222	2.357	510				
<i>PublicToPrivate</i>	0.292	0	0.455	510				
<i>ClubDeal</i>	0.310	0	0.463	510				
<i>BankSponsor</i>	0.0490	0	0.216	510				
<i>Opaque</i>	0.884	1	0.320	510				
<i>N_BankFinancing</i>	2.582	2	1.639	510				
<i>N_Dividend</i>	0.233	0	0.680	510				
<i>N_BondIssue</i>	0.202	0	0.755	510				
<i>New_financing</i>					0.0947	0	0.293	8,518
<i>Dividend</i>					0.0130	0	0.113	8,518
<i>Issue_bond</i>					0.0121	0	0.109	8,518
<i>Assets</i>	2,640	1,184	4,540	59	2,434	1,391	2,546	1,071
<i>Debt_TA</i>	0.562	0.548	0.190	59	0.811	0.685	0.465	1,061
<i>Debt_EV</i>	0.636	0.620	0.210	510	0.792	0.700	0.474	8,518
<i>Industry_DA</i>	0.206	0.213	0.1000	509	0.224	0.224	0.107	8,500
<i>IndFailure</i>	0.0640	0.0303	0.0901	510	0.0735	0	0.126	12,324
<i>SponsorFailure</i>	0.0739	0	0.166	510	0.0708	0	0.181	12,324
<i>SponsorNoHist</i>	0.208	0	0.383	510	0.327	0	0.469	12,324
<i>Borrowing_cost_all</i>	7.043	6.990	1.677	404	6.164	5.951	1.820	6,552
<i>Spread</i>	4.171	4	1.450	407	3.971	3.750	1.373	6,758
<i>SpreadT</i>	4.286	4.150	1.415	466	4.105	3.898	1.425	7,710
<i>SpreadR</i>	3.492	3.250	1.382	427	3.333	3.250	1.301	7,028
<i>Revolver</i>	0.182	0.129	0.191	510	0.166	0.117	0.184	8,518
<i>Maturity</i>	6.746	7	1.201	510	6.342	6.667	1.411	8,518
<i>MaturityT</i>	6.493	7	0.924	484	6.016	6.060	1.098	8,000
<i>MaturityR</i>	5.153	5	0.864	489	4.760	5	1.072	7,884
<i>RelatedLender</i>	0.0123	0	0.0612	510	0.0117	0	0.0634	8,518
<i>RelationshipLenderS</i>	0.143	0	0.221	510	0.181	0.0455	0.257	8,518
<i>RelationshipLenderC</i>	0.136	0	0.238	510	0.256	0	0.339	8,518
<i>P_tightening</i>	−0.745	−5.500	18.30	507	3.939	−5.300	23.88	12,312
<i>SponsorIPO</i>	0.0360	0	0.124	510	0.0255	0	0.116	12,324

Notes. Columns (1)–(4) present variables measured at deal initiation. Columns (5)–(8) present variables measured in subsequent quarters. Variable definitions appear in Table 2.

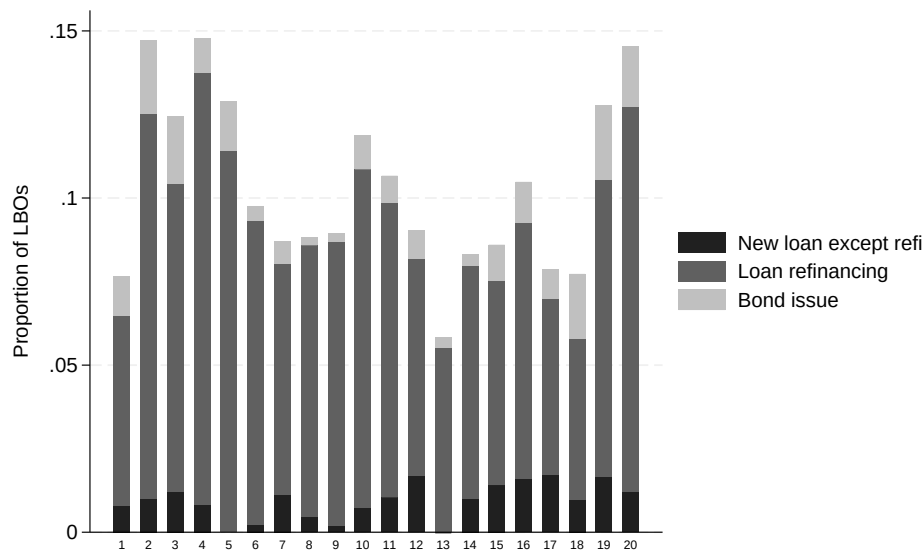
debt averages 6.746 years, and the weighted average annual borrowing cost across the initial debt structure is 7.043%. We compute this cost by adding base rates to all-in-drawn spreads where applicable and including bond coupon rates, and we weight each rate by the amount borrowed. Other measures of borrowing cost that we compute are weighted average loan *Spread*, which considers only loans expressed as spreads over Libor, and *SpreadR* and *SpreadT*, which consider only revolving credit or term loans. Sample sizes differ based on which loans have cost data available in DealScan. Additional financing events either refinance these initial facilities or add additional loans for acquisitions or dividends. Table 3 shows that there are, on average, 2.582 bank financings per deal, including the initial package, 0.233 dividend issues, and 0.202 bond issues per deal.

Figure 1 presents an average timeline of financing events by quarter of LBO life. Bond issues are a small

part of these deals. This contrasts with older deals in the time of Drexel Burnham Lambert when bonds were the bulk of LBO financing.²⁰ Figure 2 subdivides the debt packages by revolving versus term loans. This figure shows that revolving loans remain a steady proportion of the capital structure, whereas term loans increase in size over time.

Columns (5)–(8) of Table 3 present summary statistics of variables after the initial deal quarter. On one hand, one might expect that deal terms ease, conditional on the survival of the borrower until the time of refinancing. On the other hand, if the most successful deals exit prior to the maturity of their debt, the remaining troubled deals would have higher refinancing costs. We find that the average level of debt scaled by the initial EV increases over time, conditional on deal survival. This is no doubt partly because total assets grow over time. In the subsample of LBOs where total assets is available from Capital IQ; however, the debt-to-asset

Figure 1. Issue Frequency by Quarter from Deal



ratio increases as well, from an average of 0.562 at inception to 0.811 during the life of the deal. The table also shows that all four measures of borrowing cost are lower in subsequent quarters than they were in the initial quarter. This suggests that intermediate financing activity helps to decrease borrowing costs, as we confirm when we examine changes in these variables.

Table 3 also presents summary statistics on the variables measuring relationships and reputation. *SponsorFailure* averages 0.0708 during the life of the deal (column (5)) with a standard deviation of 0.181 (column (6)). In contrast, recent IPOs are less common, with a mean of 0.0255 during the life of the deal and a standard deviation of 0.116. In 32.7% of quarters during the life of the deal, there is no sponsor exit within the last eight quarters. The variables measuring relationships are

similar in magnitude. *RelationshipLenderC*, measuring a lending relationship with the company, averages 0.136 at the onset and 0.256 during the life of the deal. This reflects the fact that lenders often stay throughout the deal life. *RelationshipLenderS* presents a similar pattern, averaging 0.143 at inception and 0.181 during the life of the deal.

3.2. Autocorrelation of *SponsorFailure*

We next examine whether the findings of Braun et al. (2017) that poor outcomes are persistent within sponsor hold for our measure and within our data. Because *SponsorFailure* is the proportion of poor outcomes relative to total exits in quarters $q - 8$ through $q - 1$, we create a lagged version that examines quarters $q - 18$ through $q - 9$ and a further lag that uses the eight

Figure 2. (Color online) Debt Types Outstanding over First 20 Quarters of LBO, Scaled by Initial Deal Size (EV)

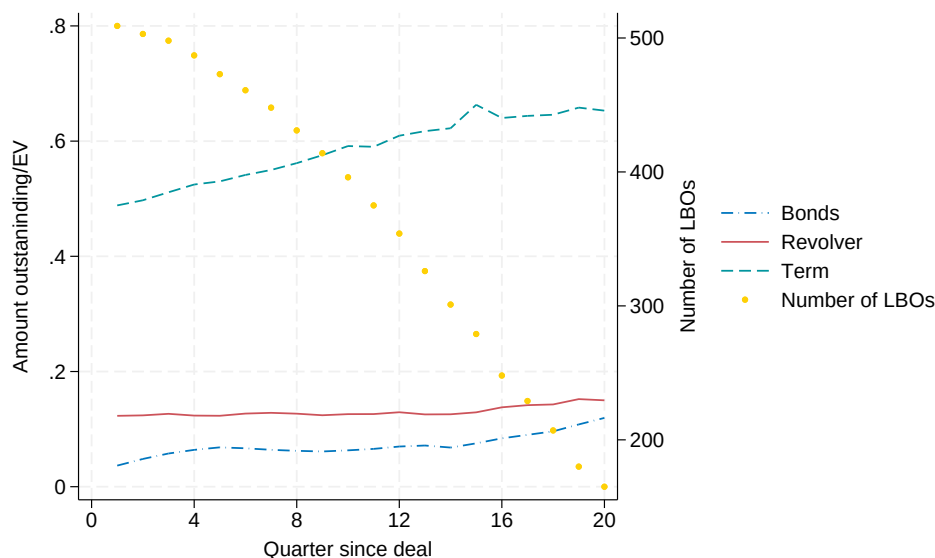
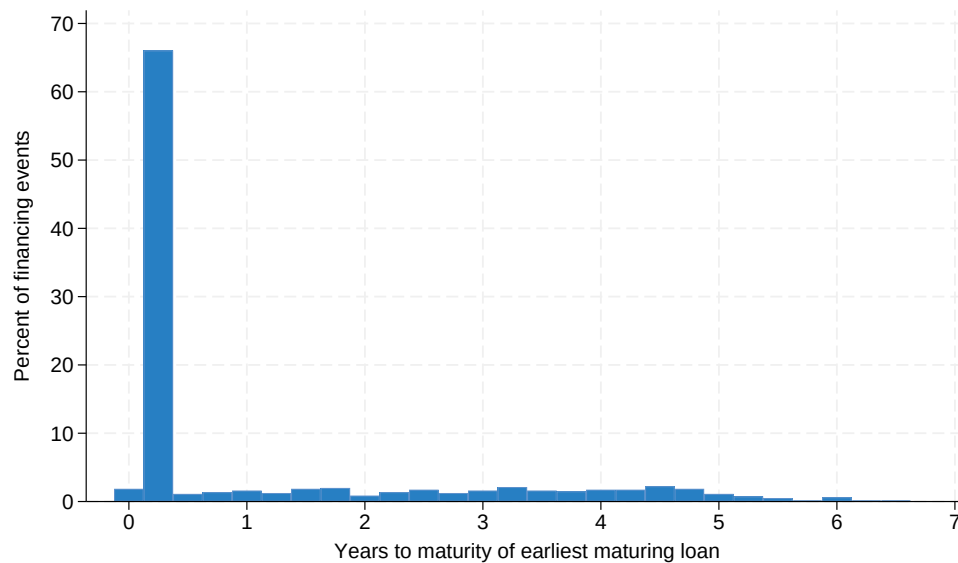


Figure 3. (Color online) Quarter of Financing Activity During the Life of the LBO by Quarter of Earliest Maturing Loan

quarters prior to that. We retain observations from quarter 0 and in multiples of eight. Table 4 presents the results with and without time fixed effects. We find that *SponsorFailure* is persistent with a coefficient of 0.077–0.125. The coefficient is not close to one, suggesting that sponsor reputation evolves quickly, and its level at the time of the buyout is not a strong signal of what it may be like years later at the deal exit. In column (3), an additional eight-quarter lag of *SponsorFailure* is not statistically related to *SponsorFailure*. The short-term nature of the predictability suggests that lenders have reason to monitor sponsors over time.

3.3. Timing of Financing Events

We now turn to summary statistics on financing events during the LBO's life. Figure 3 shows that 67% of

intermediate financing events occur exactly one quarter prior to the maturity of the earliest maturing loan. This financing activity can best be thought of as necessary, with the timing not dictated by the sponsor or lender, at least in the short term, but arguably in the long term because average maturity at deal inception displays very little variation. Table 3 shows that initial term loan maturity has a standard deviation less than one year across deals. Although it is true that a loan refinancing occurs on time because it did not occur early, it is unclear whether the decision not to refinance early should be related to the recent failure of other portfolio companies in the sponsor's portfolio other than through the channel of sponsor reputation. Because of their potentially differing motivations, we will often examine *On Time* and *Early* deals separately.

Table 4. Short-Term Persistence of Poor Deal Outcomes

Variables	(1) <i>SponsorFailure</i>	(2) <i>SponsorFailure</i>	(3) <i>SponsorFailure</i>
<i>lag8SponsorFailure</i>	0.0772* (0.08)	0.125** (0.02)	
<i>lag16SponsorFailure</i>			0.0179 (0.58)
Observations	835	835	405
R^2	0.005	0.230	0.337
Year-quarter fixed effects	No	Yes	Yes

Notes. OLS regressions of *SponsorFailure*, which is the proportion of sponsor deal exits that were write-offs, restructurings, and recapitalizations in the last eight quarters, on its lag. *SponsorFailure* is calculated using exits in quarters $q - 1$ to $q - 8$, *lag8SponsorFailure* is calculated using exits in quarters $q - 9$ to $q - 16$, and *lag16SponsorFailure* uses quarters $q - 17$ to $q - 24$. To avoid mechanical autocorrelation due to the construction of *SponsorFailure* over eight lagged quarters, we only keep observations from quarter 0 and multiples of eight. Standard errors are clustered by quarter and by Fama-French 49 industry, and p -values appear under the coefficients.

*, ** and ***Significance levels of 10%, 5%, and 1%, respectively.

Table 5. Summary Statistics for Changes at New Debt Package

Variables	(1)	(2)	(3)	(4)	(5)		(7)		(9)
	Mean	p50	Standard deviation	N	On time		Early		(5)–(7)
					Mean	N	Mean	N	t
% Δ TotalDebt	0.216***	0.104	0.356	805	0.144	557	0.379	248	***
Δ Debt_TA	0.0873***	0.0393	0.186	102	0.0593	68	0.143	34	**
Δ Borrowing_cost	−0.0375	−0.0656	0.794	566	−0.111	390	0.125	176	***
Δ Spread	−0.0779***	−0.0291	0.728	595	−0.117	409	0.00823	186	*
Δ SpreadR	−0.0216	0	0.585	649	−0.0359	466	0.0148	183	
Δ SpreadT	−0.0848***	0	0.851	748	−0.133	527	0.0298	221	**
Δ Revoloverp	−0.00369**	−0.00361	0.0528	805	−0.00472	557	−0.00138	248	
Δ Maturity	−0.369***	−0.333	1.068	805	−0.611	557	0.175	248	***
% Δ Assets	0.281***	0.0285	0.674	120	0.260	78	0.322	42	
SponsorFailure	0.0801	0	0.177	805	0.0801	557	0.0802	248	
SponsorIPO	0.0292	0	0.124	805	0.0315	557	0.0238	248	
SponsorNoHist	0.234	0	0.423	805	0.241	557	0.218	248	
P_tightening	−5.148	−8.500	14.93	805	−4.548	557	−6.495	248	*
Issue_bond	0.0311	0	0.174	805	0.0287	557	0.0363	248	
Dividend	0.138	0	0.345	805	0.0862	557	0.254	248	***
MnA	0.112	0	0.315	805	0.111	557	0.113	248	

Notes. In column (1), stars represent the statistical significance of variables that are changes. The last column presents significance levels for the *t*-test for differences in means of variables between early and on-time financings. Changes and percentage changes are winsorized at the 1% level. Variable definitions appear in Table 2.

*, ** and ***Significance levels of 10%, 5%, and 1%, respectively.

3.4. Changes at the Time of a Financing Event

Table 5 presents summary statistics for changes that occur at the time of an intermediate financing event. All changes and percentage changes are compared with the last quarter prior to the refinancing, and all change and percentage measures are winsorized at the 1% level. Several new facts emerge from this table. First, the average financing event results in a mean (median) increase in debt of 21.6% (10.4%). Of the 102 financing events for which we can compute the change, debt increases by an average (median) of 8.73% (3.93%) of total assets. Thus, these transactions are, on average, leverage increasing. The second new fact in this table is that the total cost of servicing the debt decreases, driven by an average 8.48-basis-point decrease in term loan spreads.

Columns (5)–(8) of Table 5 present the summary statistics broken down by whether the financing is *On Time*, one quarter prior to maturity of the existing debt, or *Early*, before that. The last column presents significance levels for the difference in means of the variables between the two samples. There are several notable differences. Relative to *On Time* financings, *Early* financings tend to be associated with a greater increase in debt, 37.9% versus 14.4%, and an increase in borrowing cost of 12.5 basis points rather than a decrease of 11.1 basis points. *Early* financings are much more likely to involve a dividend: 25.4% of these events involve dividends versus 8.6% of on-time financings but equally likely to be associated with an M&A event. *SponsorFailure*, *SponsorIPO*, and *SponsorNoHist* are similar across samples, which suggests that they are not strong

drivers of *Early* financing activity. The next section investigates these relations using a more formal analysis.

4. Recent Sponsor Failures, Initial Capital Structure, and New Financing Activity

4.1. Initial Capital Structure

Table 6 presents regression models of characteristics of the initial LBO debt package, which has been the principal focus of prior work. Although untabulated regressions show that macroeconomic measures such as those in Axelson et al. (2013) continue to be related to the initial capital structure of LBOs, these are subsumed by time fixed effects. We also include Fama-French 49 industry fixed effects and cluster the standard errors by quarter and by industry.

We do not find that recent sponsor failures or successes are associated with initial leverage, maturity, or spreads. Notably, we find that the measure closest to the inverse of that used in Demiroglu and James (2010), *SponsorNoHist*, is positively related to initial financing cost, suggesting that their result survives in the later sample period.

There are several possible reasons why there is no effect of *SponsorFailure* in the initial buyout debt packages. Firstly, Table 4 shows that *SponsorFailure* is predictable only two years into the future, whereas most LBOs have at least a five-year horizon and most loans have an initial maturity of over six years. In untabulated regressions, we find that, for sponsors who (perhaps know that they) will default on a loan in the next two years, initial LBO capital structure choices and costs are unchanged. Secondly, competition among lenders to

Table 6. Initial LBO Leverage and Cost and Sponsor Reputation

Variables	(1) Debt/EV	(2) Borrowing cost	(3) Borrowing cost public to private	(4) Loan spread	(5) Revolver spread	(6) Term spread
<i>SponsorFailure</i>	−0.00243 (0.97)	0.0805 (0.86)	0.650 (0.41)	−0.0811 (0.82)	−0.0384 (0.93)	−0.285 (0.30)
<i>SponsorIPO</i>	−0.00255 (0.98)	0.574 (0.33)	1.315 (0.45)	0.508 (0.38)	−0.00958 (0.97)	0.0756 (0.82)
<i>SponsorNoHist</i>	−0.0178 (0.57)	0.308* (0.10)	0.962** (0.03)	0.230 (0.15)	0.185 (0.19)	0.0167 (0.92)
<i>PublicToPrivate</i>	0.0187 (0.41)	0.0455 (0.82)		0.173 (0.36)	0.0961 (0.37)	0.147 (0.41)
<i>ClubDeal</i>	−0.0263 (0.24)	0.0332 (0.80)	−0.379 (0.44)	−0.125 (0.38)	0.118 (0.44)	−0.130 (0.35)
<i>BankSponsor</i>	0.0126 (0.79)	−0.149 (0.63)	−0.288 (0.57)	−0.213 (0.44)	0.110 (0.61)	−0.147 (0.59)
<i>RelationshipLenderC</i>	0.0163 (0.66)	−0.123 (0.66)	0.450 (0.34)	−0.120 (0.67)	−0.381* (0.07)	0.128 (0.55)
<i>RelationshipLenderS</i>	−0.0201 (0.60)	0.0341 (0.91)	−1.096** (0.02)	−0.0682 (0.80)	0.236 (0.27)	0.230 (0.41)
<i>RelatedLender</i>	0.0665 (0.72)	−0.929* (0.09)	−0.444 (0.53)	−0.500 (0.40)	0.849 (0.17)	−1.867*** (0.00)
<i>Opaque</i>	−0.0808* (0.07)	−0.493* (0.07)	−0.628 (0.18)	−0.152 (0.56)	−0.144 (0.56)	−0.0218 (0.93)
<i>Debt_EV</i>		0.761* (0.05)	0.986 (0.32)	0.145 (0.64)	−0.435 (0.21)	0.0133 (0.97)
<i>logEV</i>	−0.00910 (0.55)	−0.339*** (0.00)	−0.403 (0.33)	−0.386*** (0.00)	−0.499*** (0.00)	−0.525*** (0.00)
<i>IndFailure</i>	0.117 (0.40)	0.208 (0.71)	−1.989 (0.27)	0.0339 (0.95)	0.986 (0.17)	0.179 (0.78)
<i>Industry_DA</i>	0.362 (0.25)	0.849 (0.68)	16.27*** (0.00)	0.472 (0.79)	2.760* (0.08)	−0.0280 (0.99)
Observations	509	403	120	406	426	465
R ²	0.387	0.755	0.919	0.698	0.717	0.631
FF49 fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes. The dependent variable appears in the column header. Standard errors are clustered by time in quarters and by Fama-French 49 industry, and *p*-values appear under the coefficients. Variable definitions appear in Table 2.

*, ** and ***Significance levels of 10%, 5%, and 1%, respectively.

secure the initial LBO deal financing among these frequent deal sponsors may constrain terms (anecdotally, proposed terms are very similar across lenders). In untabulated results, the similarity of lead lenders across refinancing events in our sample averages more than 90%, so borrowers do not seem to be inclined to seek new lenders after the initial LBO financing. Less competition among lenders at refinancing time leaves more latitude for changes in terms. Thirdly, the lack of a result of initial debt could also be because some LBO deals exit prior to needing to refinance. Although Table 3 shows that the median firm has financing activity after the buyout, many firms do not. In untabulated regressions on the subsample of LBO firms that have financing events after the initial financing, the initial LBO deal parameters of these firms are still not related to *SponsorFailure*. In Section 6, we find that the change in debt terms at the time of a new financing event is related to the probability of failure of the deal, which lends further credence to the first two explanations.

Table 6 does not show a strong role for our measures of relationships. Although this finding appears to contrast with results in Ivashina and Kovner (2011), that paper uses a much earlier sample period, only public-to-private deals, and a different measure, as discussed earlier. In Table 6, column (3), we do find that *RelationshipLenderS* is associated with lower initial borrowing cost for our subsample of public-to-private deals. We also do not find that bank sponsors obtain better loan terms in our sample. However, in untabulated regressions, we find that these sponsors do obtain better terms for deals with dates prior to 2010, consistent with the findings of Fang et al. (2013).

4.2. Incidence of New Financing Activity

Examining levels across LBOs makes it difficult to control for all differences between deals that are unobserved by the researcher. These characteristics are better controlled for by examining changes within each deal over time. Table 7 explores the incidence of new financing

activity, dividends, and bond issues over the time that a sponsor holds the LBO target. We distinguish between *On Time* (in the quarter of, or the before, the earliest loan maturity in the current package) and *Early* (in any earlier quarter) financing events using interaction terms with *SponsorFailure*, because *On Time* financing activity is pre-determined to a great extent by the maturity date of the prior loan package. The dependent variable is an

indicator for whether there is a new financing event in a given LBO quarter, and the sample is all quarters after the initial financing until exit or until the end of the sample period for deals that have not yet exited. Because Ai and Norton (2003) question the interpretation of interaction terms in Probit models, we also tabulate the linear model coefficients of greatest interest in the last lines of Table 7.

Table 7. Determinants of Financing Activity After the LBO

Variables	(1) Financing event	(2) Financing event	(3) Financing event banks looser	(4) Financing event banks tighter	(5) Dividend	(6) Dividend	(7) Dividend banks looser	(8) Dividend banks tighter
<i>SponsorFailure</i>	−0.0941 (0.51)				−0.508* (0.08)			
<i>SponsorFailure</i> × <i>OnTime</i>		−0.238 (0.26)	−0.0727 (0.80)	−0.375 (0.34)		−1.358** (0.03)	−0.468 (0.38)	−5.591*** (0.01)
<i>SponsorFailure</i> × <i>Early</i>		−0.00686 (0.96)	−0.382 (0.13)	0.359* (0.08)		−0.226 (0.40)	−0.146 (0.70)	−1.273 (0.12)
<i>SponsorIPO</i>	−0.198 (0.38)	−0.197 (0.38)	0.0759 (0.79)	−0.526 (0.19)	−0.152 (0.65)	−0.156 (0.64)	0.213 (0.67)	−0.679 (0.19)
<i>SponsorNoHist</i>	−0.152** (0.02)	−0.154** (0.02)	−0.158 (0.11)	−0.0972 (0.38)	−0.112 (0.29)	−0.117 (0.28)	−0.0867 (0.55)	−0.0661 (0.77)
<i>Early</i>	−1.525*** (0.00)	−1.544*** (0.00)	−1.458*** (0.00)	−1.703*** (0.00)	−0.373*** (0.00)	−0.435*** (0.00)	−0.176 (0.18)	−0.961*** (0.00)
<i>PublicToPrivate</i>	−0.0965 (0.11)	−0.0972 (0.11)	−0.0779 (0.31)	−0.138 (0.19)	−0.0605 (0.54)	−0.0659 (0.51)	−0.0374 (0.80)	−0.0505 (0.83)
<i>ClubDeal</i>	0.0571 (0.31)	0.0566 (0.32)	0.0126 (0.89)	0.0793 (0.25)	0.148 (0.19)	0.142 (0.21)	0.00578 (0.97)	0.435** (0.03)
<i>BankSponsor</i>	−0.0836 (0.50)	−0.0845 (0.49)	−0.0978 (0.56)	−0.0894 (0.62)	−0.348 (0.20)	−0.369 (0.17)	−0.170 (0.57)	−7.749*** (0.00)
<i>lagRelatedLender</i>	−0.147 (0.79)	−0.144 (0.79)	0.0235 (0.97)	−0.595 (0.42)	0.437 (0.41)	0.486 (0.36)	0.275 (0.67)	3.569 (0.13)
<i>lagRelationshipLenderC</i>	−0.0359 (0.66)	−0.0383 (0.64)	0.00850 (0.95)	−0.0412 (0.75)	0.170 (0.25)	0.163 (0.27)	0.347 (0.13)	−0.0768 (0.77)
<i>lagRelationshipLenderS</i>	−0.0964 (0.37)	−0.0952 (0.38)	−0.0462 (0.76)	−0.182 (0.27)	−0.153 (0.46)	−0.145 (0.49)	−0.202 (0.45)	−0.00541 (0.99)
<i>Opaque</i>	−0.133* (0.07)	−0.132* (0.07)	−0.120 (0.32)	−0.165 (0.12)	0.103 (0.51)	0.0979 (0.53)	0.0631 (0.77)	−0.0566 (0.84)
<i>lagDebt_EV</i>	−0.0923 (0.10)	−0.0927 (0.10)	−0.0378 (0.63)	−0.118 (0.14)	0.0964 (0.15)	0.0935 (0.17)	0.0962 (0.26)	0.225* (0.06)
<i>logEV</i>	0.0911*** (0.01)	0.0914*** (0.01)	0.218*** (0.00)	−0.00780 (0.88)	0.00528 (0.92)	0.00526 (0.92)	0.135 (0.10)	−0.102 (0.22)
<i>IndFailure</i>	0.0663 (0.71)	0.0680 (0.70)	0.339 (0.17)	−0.0812 (0.74)	0.552 (0.11)	0.557 (0.12)	0.956** (0.05)	0.0860 (0.89)
Observations	8,411	8,411	3,438	4,973	8,411	8,411	3,438	4,973
R^2	0.314	0.314	0.327	0.333	0.238	0.240	0.282	0.372
FF49 fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Linear coefficient <i>SponsorFailure</i>	−0.006 (0.70)				−0.011* (0.06)			
Linear coefficient <i>SponsorFailure</i> × <i>OnTime</i>		−0.040 (0.52)	−0.005 (0.95)	−0.100 (0.29)		−0.052 (0.00)	−0.029 (0.22)	−0.094 (0.00)
Linear coefficient <i>SponsorFailure</i> × <i>Early</i>		0.000 (0.97)	−0.034** (0.05)	0.036*** (0.01)		−0.005 (0.36)	−0.002 (0.88)	−0.006 (0.19)

Notes. Probit models of early refinancing activity and of dividend issuance, merger and acquisition activity, and bond issuance. Variable definitions appear in Table 2. Standard errors are clustered by time in quarters for probit models and by time in quarters and by Fama-French 49 industry for linear models. Linear model coefficients on *SponsorFailure* and its p -value are listed in the last rows of the table. p -values appear under the coefficients.

*, **, and ***Significance levels of 10%, 5%, and 1%, respectively.

In this table, the negative coefficient on *Early* reflects that financing deals happen much less often in these quarters. Using the delta method,²¹ the probit model in column (1) with the coefficient of -1.525 on *Early* implies that moving from *Early* to *On-time* increases the odds of a financing event by 0.33, net of the effects of the other variables. Table 3, column (5), shows that financing events occur in 9.47% of postdeal quarters. In untabulated results, this is 3.5% of *Early* quarters and 41.5% of *On Time* quarters.

Columns (1)–(3) of Table 7 show that the lack of recent sponsor exits, *SponsorNoHist*, is significantly negatively related to the incidence of financing activity. Using the delta method, the coefficient in column (1) implies that a sponsor with no history of exits in the last two years has a 0.0126 lower probability of undertaking a financing out of a baseline probability of a financing event of 0.0947 from Table 3. Columns (3) and (4) divide the sample into periods when banks are tightening more compared with the last financing date, which could be the inception of the LBO, and when they are loosening more. In an untabulated robustness test, we find that results are similar whether we use this measure or, instead, the difference in the high-yield rate since the last financing event. These two variables are more than 50% correlated.

Columns (1)–(4) of Table 7 show that *SponsorFailure* is related to the incidence of a financing event, but only in *Early* periods when banks are tightening more than they were at the last financing event. In column (4), there is a positive effect of *SponsorFailure* $\times$ *Early*, and its significance is confirmed at the bottom of the table using the linear model. A one-standard-deviation increase in *SponsorFailure*, 0.181 from Table 3, column (5), is associated with an 0.0065 increase in the incidence of financing activity in an early quarter. Recall that 3.5% of *Early* quarters have financing events. Many of these events may be initiated by lenders, perhaps due to violated covenants.

Columns (5)–(8) of Table 7 examine the incidence of debt financing events that include dividends. Using the delta method, we find that a one-standard-deviation increase in *SponsorFailure* is associated with a 0.0025 lower incidence of financing that includes a dividend issuance in a given quarter, compared with a baseline issuance of 0.0130 from Table 3. Thus, sponsors who have recently failed are 19.2% less likely to issue dividends in their current portfolio companies. Column (6) shows that this effect is only significant in *On Time* quarters and columns (7) and (8) show that this occurs when banks are tightening more compared with the time of the borrower's last financing event. In untabulated results, we do not find a strong effect of *SponsorFailure* on M&A activity or on bond issuance.

In Table 8, we examine all financing events and consider whether they constitute a loosening or tightening

of lending standards. We define a loosening to be an improvement in loan size, loan cost, or maturity while the other variables improve or stay the same. Similarly, a tightening occurs when at least one of the loan size, loan cost, or loan maturity deteriorates in the new financing arrangement, whereas the others, at best, stay the same. One caveat is that Table 5 shows that new financing deals during the life of the LBO often use a shorter maturity, likely due to a closer expected exit date. This is particularly true for *On Time* deals. Thus, our definition may miss some loosening events and mis-categorize some mixed events as tightening. Out of 595 financing events where the necessary variables are not missing, 49% are loosening, 6.7% are tightening, and the remainder are mixed.

Columns (1) and (4) of Table 8 present Probit models for a loosening and tightening event given a financing event. Equivalent linear coefficients for key variables appear in the last lines of the table. As behavior might depend on economic conditions, columns (3) and (4) divide the sample into times when banks are loosening their to a greater extent since the firm's last financing event, which could be the inception of the LBO.

Table 8, column (1), shows that *SponsorFailure* is negatively associated with financing events that loosen standards for LBO firms. Using the delta method, a one-standard-deviation increase in *SponsorFailure* is associated with a six-percentage-point lower probability of a loosening financing compared with the mean probability of 49% conditional on a financing event. Columns (2), (3), and (4) show that this result is strongest in periods when banks are loosening on average relative to the time of the last financing event, which could have been the LBO inception. This result is also strongest for *On Time* financing deals, where the timing is less likely to be endogenous to the borrower's financial situation. Thus, *SponsorFailure* is negatively associated with the ability of sponsors to obtain better financing deals when conditions have improved since the last financing event. Columns (5) and (6) investigate the incidence of tightening events and find that they are positively related to *SponsorFailure*, again for *On Time* cases. There are not enough tightening financing events to present the equivalent regressions separately.

4.3. Recent Sponsor Failures and Changes in Leverage

Table 9 examines the changes in the amount of debt at the time of the new financing event. Predictably, deals that include a *Dividend* or M&A or are *Early* are associated with larger percentage increases in debt. Column (1) shows that, overall, *SponsorFailure* is negatively related to the change in debt, with a coefficient of 0.112. A one-standard-deviation increase in *SponsorFailure*, 0.181, is associated with a two-percentage-point lower increase in debt at the time of the refinancing. Recall

Table 8. Probit Models of Loosening and Tightening Loan Terms Conditional on a Financing Event

Variables	(1) Loosening	(2) Loosening	(3) Loosening banks looser	(4) Loosening banks tighter	(5) Tightening	(6) Tightening
<i>SponsorFailure</i>	−1.127** (0.01)				1.301 (0.11)	
<i>SponsorFailure</i> × <i>OnTime</i>		−1.225** (0.02)	−2.061*** (0.00)	−0.430 (0.73)		2.934** (0.03)
<i>SponsorFailure</i> × <i>Early</i>		−0.847 (0.22)	−6.688*** (0.00)	1.246 (0.32)		−0.208 (0.83)
<i>SponsorIPO</i>	−0.887 (0.27)	−0.890 (0.27)	−1.540 (0.16)	2.345 (0.31)	−3.370*** (0.00)	−3.706*** (0.00)
<i>SponsorNoHist</i>	0.0210 (0.91)	0.0213 (0.91)	−0.0495 (0.87)	0.149 (0.65)	0.0124 (0.98)	0.160 (0.77)
<i>Early</i>	−0.190 (0.31)	−0.215 (0.28)	0.274 (0.38)	−0.310 (0.28)	0.459 (0.43)	0.873* (0.09)
<i>lagRelatedLender</i>	3.332** (0.01)	3.368** (0.01)	4.194 (0.17)	2.873 (0.50)	−21.60** (0.04)	−25.21** (0.01)
<i>lagRelationshipLenderC</i>	0.239 (0.31)	0.237 (0.32)	0.264 (0.53)	0.221 (0.50)	−1.146** (0.03)	−1.148** (0.04)
<i>lagRelationshipLenderS</i>	−0.628* (0.06)	−0.627* (0.06)	−0.411 (0.38)	−1.084* (0.06)	1.225 (0.16)	1.439 (0.11)
Observations	595	595	312	283	595	595
R^2	0.245	0.245	0.371	0.375	0.618	0.630
Controls	Yes	Yes	Yes	Yes	Yes	Yes
FF49 and year-quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Linear coefficient <i>SponsorFailure</i>	−0.319				0.101	
Linear p <i>SponsorFailure</i>	0.0100				0.230	
Linear coefficient <i>SponsorFailure</i> × <i>OnTime</i>		−0.375** (0.02)	−0.546*** (0.00)	−0.162 (0.72)		0.121 (0.12)
Linear coefficient <i>SponsorFailure</i> × <i>Early</i>		−0.214 (0.18)	−0.840*** (0.00)	0.0940 (0.67)		0.0630 (0.62)

Notes. A loosening event is defined as a relaxing of at least one of the spread, maturity or size of the loan, while the others do not deteriorate. A tightening is defined as a tightening of either the interest rate, maturity, or size of the loan, while the other measures remain at best constant. Loosening and tightening financing types are defined only for the events where financing cost information is not missing, and of these, 49% are loosening, 6.7% are tightening, and the rest are ambiguous. Control variables not shown are *PublicToPrivate*, *BankSponsor*, *ClubDeal*, *Opaque*, *lagDebt_EV* *logEV*, *IndFailure*. Standard errors are clustered by time in quarters and by Fama-French 49 industry, and p -values appear under the coefficients. Variable definitions appear in Table 2.

*, ** and ***Significance levels of 10%, 5%, and 1%, respectively.

from Table 5 that the average change in debt is 21.6% at the time of the financing (14.4% for *OnTime* events and 37.9% for *Early* events). Column (2) shows that the result in column (1) is driven by *Early* events. Columns (3) and (4) show that this effect is stronger when banks are loosening lending standards relative to the time of the borrower's last financing event. Thus, sponsors that have recently failed find it more difficult to borrow in an unscheduled manner and potentially pass up new business opportunities.

4.4. Recent Sponsor Failures and the Cost of New and Refinanced Debt

Table 10 examines the change in various measures of the cost of debt at the time of a financing event. The first two measures, $\Delta Cost$ and $\Delta Spread$, are comprehensive measures of the total cost and spread of the debt package and require all data to exist, resulting in smaller sample sizes. The latter two measures, $\Delta SpreadR$ and $\Delta SpreadT$, require only those spread data to exist, respectively. The

first two columns examine $\Delta Cost$, the weighted average annual cost of debt, including bonds. In the case of variable rate debt, we add the relevant base interest rate (generally Libor) to the spread. Columns (3) and (4) present the weighted average spread on loans, and the remaining columns examine revolving debt and term debt separately. In untabulated regressions, we find similar results when we restrict the sample to spreads over Libor (we mainly remove loans based on the Secured Overnight Financing Rate).

Columns (1) and (2) show that *SponsorFailure* is positively related to the change in total cost at the time of new financing, in particular, for *Early* financings. A one-standard-deviation increase in *SponsorFailure*, 0.181 from Table 3, column (5), is associated with a 9.5-basis-point increase in cost, driven by a 23.9-basis-point increase in cost at refinancing using the coefficient in column (2) on *Early* financings. In column (3), the overall spread is significantly related to *SponsorFailure*, and a one-standard-deviation increase in *SponsorFailure* is

Table 9. Changes in the Amount of Debt at a New Financing Event

Variables	(1) %ΔDebt	(2) %ΔDebt	(3) %ΔDebt banks looser	(4) %ΔDebt banks looser	(5) %ΔDebt banks tighter	(6) %ΔDebt banks tighter
<i>SponsorFailure</i>	−0.112* (0.06)		−0.107 (0.23)		−0.113 (0.31)	
<i>SponsorFailure</i> × <i>OnTime</i>		−0.0533 (0.35)		−0.0379 (0.66)		−0.102 (0.43)
<i>SponsorFailure</i> × <i>Early</i>		−0.233** (0.02)		−0.306*** (0.00)		−0.127 (0.52)
<i>SponsorIPO</i>	0.0613 (0.50)	0.0625 (0.49)	0.0637 (0.48)	0.0674 (0.46)	0.0204 (0.94)	0.0214 (0.93)
<i>SponsorNoHist</i>	−0.0565* (0.06)	−0.0550* (0.06)	−0.0596 (0.26)	−0.0560 (0.30)	−0.108** (0.05)	−0.108** (0.05)
<i>lagRelationshipLenderC</i>	−0.0391 (0.46)	−0.0364 (0.49)	−0.0369 (0.54)	−0.0245 (0.69)	−0.0581 (0.38)	−0.0587 (0.37)
<i>lagRelationshipLenderS</i>	0.0192 (0.79)	0.0174 (0.81)	0.0184 (0.83)	0.00912 (0.92)	0.00577 (0.95)	0.00624 (0.95)
<i>lagRelatedLender</i>	0.00336 (0.99)	−0.00646 (0.97)	−0.101 (0.74)	−0.124 (0.68)	0.0843 (0.71)	0.0848 (0.71)
<i>Early</i>	0.212*** (0.00)	0.225*** (0.00)	0.196*** (0.00)	0.216*** (0.00)	0.251*** (0.00)	0.253*** (0.00)
<i>MnA</i>	0.282*** (0.00)	0.284*** (0.00)	0.314*** (0.00)	0.312*** (0.00)	0.218*** (0.00)	0.219*** (0.00)
<i>Dividend</i>	0.188*** (0.00)	0.188*** (0.00)	0.190*** (0.00)	0.190*** (0.00)	0.173** (0.03)	0.173** (0.03)
<i>lagDebt_EV</i>	−0.159*** (0.00)	−0.160*** (0.00)	−0.126*** (0.01)	−0.127*** (0.01)	−0.199*** (0.00)	−0.199*** (0.00)
<i>logEV</i>	−0.0780*** (0.00)	−0.0783*** (0.00)	−0.0851*** (0.00)	−0.0861*** (0.00)	−0.0843*** (0.00)	−0.0844*** (0.00)
<i>BankSponsor</i>	0.0160 (0.68)	0.0184 (0.65)	0.0491 (0.44)	0.0635 (0.32)	−0.0263 (0.71)	−0.0263 (0.71)
<i>PublicToPrivate</i>	0.00149 (0.96)	0.00117 (0.97)	0.00945 (0.86)	0.00937 (0.86)	−0.0159 (0.73)	−0.0156 (0.74)
<i>ClubDeal</i>	0.0539** (0.02)	0.0530** (0.02)	0.0786*** (0.00)	0.0775*** (0.00)	0.0252 (0.34)	0.0253 (0.34)
<i>Opaque</i>	−0.0291 (0.30)	−0.0291 (0.31)	−0.0621 (0.23)	−0.0576 (0.26)	−0.0180 (0.73)	−0.0184 (0.71)
<i>IndFailure</i>	−0.225** (0.04)	−0.219** (0.04)	−0.134 (0.38)	−0.115 (0.45)	−0.248** (0.05)	−0.249** (0.05)
Observations	805	805	438	438	367	367
R^2	0.366	0.367	0.389	0.392	0.489	0.489
FF49 and year-quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes. OLS regressions for percentage changes in debt and changes in debt to assets. Standard errors are clustered by time in quarters and by Fama-French 49 industry, and p -values appear under the coefficients. Variable definitions appear in Table 2.

*, ** and ***Significance levels of 10%, 5%, and 1%, respectively.

associated with an 8.5-basis-point increase in spread. Column (4) shows that this is again driven by *Early* financings with a similar magnitude as a result in column (2). Columns (5) and (6) examine revolving debt only and find similar results. Columns (7) and (8) examine term loans and show that the effects on the overall cost of debt are principally driven by these. This is unsurprising, as revolving debt is less risky for the lender because it is easier for a bank to recall, especially if it is unused. Column (7) shows that a one-standard-deviation increase in *SponsorFailure* is associated with a $0.181 \times 0.679 = 12.3$ -basis-point increase in the weighted average term loan spread, and column (8) shows that this result is 7.1 basis points in *OnTime* financings and 27.3 basis points in *Early* financings. For comparison,

recall that Table 5 shows that the average change in term loan spread is −13.3 basis points for *OnTime* financings and +2.98 basis points for *Early* financings.

Columns (9) and (10) break down the sample into time periods when banks are, on the net, loosening or tightening standards relative to the time of the last financing event for the borrower. When banks are loosening more, borrowers pay about 11.6 basis points more for each standard deviation of *SponsorFailure* regardless of whether the financing was *OnTime* or *Early*. When banks are tightening more, the difference is similar at 12.1 basis points.

These findings show that *SponsorFailure* is negatively associated with the ability of sponsors to obtain better financing, even when times have improved since the

Table 10. Changes in the Cost of Debt

Variables	(1) ΔCost	(2) ΔCost	(3) ΔSpread	(4) ΔSpread	(5) ΔSpread revolver	(6) ΔSpread revolver	(7) ΔSpread term	(8) ΔSpread term	(9) ΔSpread term, banks looser	(10) ΔSpread term, banks tighter
<i>SponsorFailure</i>	0.527* (0.06)		0.468* (0.05)		0.0566 (0.69)		0.679*** (0.00)		0.639*** (0.00)	0.671* (0.08)
<i>SponsorFailure</i> × <i>OnTime</i>		0.108 (0.67)		0.0669 (0.76)		−0.197 (0.29)		0.395** (0.03)		
<i>SponsorFailure</i> × <i>Early</i>		1.318*** (0.01)		1.232*** (0.00)		0.656** (0.02)		1.510*** (0.00)		
<i>SponsorIPO</i>	0.207 (0.54)	0.196 (0.54)	0.103 (0.73)	0.101 (0.72)	0.0688 (0.71)	0.0716 (0.71)	0.228 (0.43)	0.215 (0.45)	0.516 (0.15)	−0.788 (0.29)
<i>SponsorNoHist</i>	−0.0293 (0.66)	−0.0375 (0.56)	−0.0970* (0.06)	−0.102** (0.03)	−0.0478 (0.35)	−0.0546 (0.29)	−0.0259 (0.70)	−0.0317 (0.64)	−0.00236 (0.98)	−0.0970 (0.44)
<i>lagRelationshipLenderC</i>	−0.108 (0.29)	−0.126 (0.20)	−0.0942 (0.30)	−0.108 (0.22)	−0.00393 (0.96)	−0.0132 (0.87)	0.117 (0.25)	0.0980 (0.34)	0.258* (0.07)	−0.112 (0.52)
<i>lagRelationshipLenderS</i>	0.292** (0.05)	0.279* (0.06)	0.302* (0.06)	0.293* (0.07)	0.0939 (0.32)	0.107 (0.27)	0.199 (0.35)	0.216 (0.31)	−0.0141 (0.95)	0.316 (0.38)
<i>lagRelatedLender</i>	−1.137* (0.05)	−1.043* (0.08)	−0.611 (0.22)	−0.500 (0.33)	−0.156 (0.59)	−0.142 (0.63)	0.0851 (0.81)	0.157 (0.66)	−0.162 (0.78)	0.779* (0.08)
<i>Early</i>	−0.00179 (0.99)	−0.0923 (0.46)	−0.0766 (0.49)	−0.165 (0.13)	0.0777 (0.43)	0.00974 (0.90)	−0.0312 (0.68)	−0.115 (0.12)	0.0197 (0.90)	−0.130 (0.32)
<i>MnA</i>	−0.133 (0.15)	−0.160* (0.10)	−0.0666 (0.36)	−0.0850 (0.24)	0.0667 (0.18)	0.0480 (0.36)	−0.0198 (0.87)	−0.0283 (0.82)	0.0211 (0.92)	−0.0208 (0.88)
<i>Dividend</i>	0.125 (0.37)	0.121 (0.37)	0.235 (0.11)	0.233 (0.10)	0.169 (0.11)	0.162 (0.14)	0.0929 (0.51)	0.0924 (0.51)	0.0547 (0.84)	0.0567 (0.62)
%Δ <i>TotalDebt</i>	0.318*** (0.01)	0.339*** (0.00)	0.188* (0.07)	0.207** (0.03)	−0.150 (0.14)	−0.127 (0.21)	0.193 (0.12)	0.200 (0.10)	0.174 (0.28)	0.0896 (0.68)
Δ <i>Maturity</i>	0.0468 (0.19)	0.0339 (0.36)	0.0690** (0.01)	0.0562* (0.05)	−0.0543 (0.19)	−0.0556 (0.18)	0.0657** (0.05)	0.0638* (0.05)	−0.00665 (0.91)	0.177** (0.03)
Δ <i>Revolverp</i>	−0.730 (0.22)	−0.850 (0.16)	−0.459 (0.47)	−0.586 (0.38)	−1.012* (0.09)	−0.918* (0.10)	0.989 (0.17)	0.961 (0.18)	1.316 (0.15)	0.727 (0.58)
<i>lagDebt_EV</i>	0.225** (0.02)	0.253** (0.01)	0.0976 (0.20)	0.122 (0.12)	0.0331 (0.40)	0.0364 (0.34)	0.119* (0.10)	0.119* (0.08)	0.214* (0.07)	−0.0251 (0.82)
<i>logEV</i>	−0.000506 (0.99)	0.00342 (0.93)	−0.0265 (0.49)	−0.0228 (0.57)	0.0240 (0.44)	0.0269 (0.39)	−0.0316 (0.40)	−0.0306 (0.42)	0.000463 (0.99)	−0.0856 (0.24)
<i>BankSponsor</i>	0.0747 (0.76)	0.0455 (0.84)	0.0523 (0.79)	0.0386 (0.84)	−0.0867 (0.57)	−0.0910 (0.50)	0.0142 (0.92)	−0.00601 (0.97)	0.00695 (0.97)	−0.0935 (0.69)
<i>PublicToPrivate</i>	0.147** (0.05)	0.152* (0.06)	0.132* (0.08)	0.139* (0.08)	−0.00826 (0.89)	−0.00975 (0.88)	0.135** (0.03)	0.139** (0.04)	0.101 (0.20)	0.251* (0.06)
<i>ClubDeal</i>	0.0793 (0.30)	0.0933 (0.21)	0.0648 (0.30)	0.0772 (0.22)	−0.0399 (0.19)	−0.0377 (0.24)	0.0472 (0.52)	0.0552 (0.42)	0.0768 (0.39)	−0.0207 (0.85)
<i>Opaque</i>	−0.166 (0.17)	−0.166 (0.18)	−0.167 (0.18)	−0.166 (0.19)	−0.197** (0.03)	−0.200** (0.02)	−0.150 (0.23)	−0.147 (0.25)	−0.214 (0.33)	−0.192 (0.36)
<i>IndFailure</i>	−0.255 (0.38)	−0.269 (0.38)	−0.523* (0.08)	−0.524* (0.09)	0.142 (0.67)	0.135 (0.70)	−0.505 (0.14)	−0.526 (0.12)	−0.393 (0.31)	−0.347 (0.51)
Observations	566	566	595	595	648	648	748	748	408	340
R ²	0.369	0.383	0.329	0.345	0.292	0.304	0.274	0.282	0.356	0.376
FF49 and year-quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

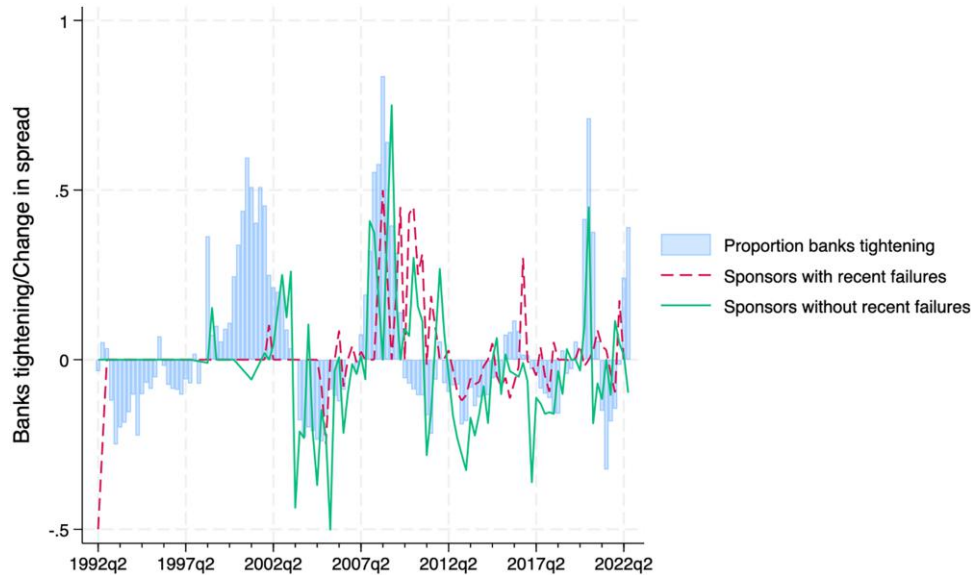
Notes. Columns (1) and (2) present weighted average total cost, and columns (3) and (4) present weighted average loan spreads. Sample sizes reflect the availability of all components of the dependent variable. The control variable *Maturity* is the weighted average for the loan type indicated in the column header. Variable definitions appear in Table 2. Standard errors are clustered by time in quarters and by Fama-French 49 industry, and *p*-values appear under the coefficients.

*, ** and ***Significance levels of 10%, 5%, and 1%, respectively.

last financing event and even when the timing of the financing event is reasonably exogenous. Figure 4 illustrates this by plotting the changes in the weighted average spread for financing activity in the quarters where spreads are available. The bars represent the net proportion of banks that are tightening, according to the

Federal Reserve. The solid line represents sponsors that have not failed in the last two years or since the latest financing of the company, and the dashed line represents sponsors that have failed in that time frame. One can see that sponsors without recent failures are better able to extract lower rates from lenders.

Figure 4. (Color online) Average Changes in the Weighted Average Cost of Financing at the Time of Financing Events by Quarter for the Companies of Sponsors with Recent Failures and for the Companies of Sponsors Without Recent Failures



Note. The bars represent the net proportion of banks tightening their lending standards in that quarter according to the Federal Reserve.

Are these changes in borrowing costs economically significant to a sponsor? For a conservatively financed leveraged buyout with 79% debt as in Table 3, column (5), an 8.5-basis-point decrease in spread can increase returns to equity by $8.5 \times 0.79 / 0.21 = 32$ basis points per year, ignoring taxes. As this effect extends to all companies or portfolios controlled by the sponsor as the need to refinance arises, it is likely to represent a sizeable amount.

4.5. Recent Sponsor Failures and Changes in Maturity

In a public firm setting, Dangl and Zechner (2021), building on Admati et al. (2018) and DeMarzo and He (2021), show that short debt maturities commit equity holders to reduce leverage when they must refinance expiring debt in low-profitability states. In these states, they choose to issue equity to refinance some of the expiring debt. Applying this to our setting, debt maturity at inception and refinancing time should be related to measures that are associated with future distress, such as *SponsorFailure*. Instead, we find in Table 3 that initial LBO debt maturity has low variation at around six years, with less than a one-year standard deviation for term loans, and Table 5 shows that refinancing maturities also show low variability. In untabulated regressions, initial maturity or its changes at refinancing time are not related to *SponsorFailure*. Thus, it does not seem that debt maturity is used as a tool by lenders to constrain sponsors.

These results suggest that the effect described in Dangl and Zechner (2021) is not first order in our setting. Although no theory captures all institutional detail,

there are some key differences between our setting and theirs. In Dangl and Zechner (2021), the firm has a perpetual life. This contrasts with private equity funds, which typically have 10- to 12-year lives. Axelson et al. (2009) explain that finite lives are a way for sponsors to build a reputation through successive funds. Moreover, in Dangl and Zechner (2021), creditors are atomistic and unable to bargain, whereas lenders have bargaining power in our setting, in particular, for refinancing events. In Dangl and Zechner (2021), equity holders are allowed to decide on the mix of debt and equity that is used to refinance debt as it comes due. In private equity, additional equity issuances are rare and would be done through an equity cure provision. These can be limited contractually by lenders and are not always welcomed because they delay lenders' ability to foreclose. Moreover, sponsors invest most of their fund's committed capital during the investment period, which is the first three years of the fund's life. There is often no uninvested capital left over after that point, and sponsors must make special arrangements with their limited partners if they need to inject more equity into a portfolio company.

5. Pre-LBO Behavior, Matched Firms, and Other Robustness Tests

One may ask whether these characteristics and results are unique to private equity ownership or are perhaps an artifact of the types of LBO targets that private equity investors tend to favor in certain time periods. To investigate this possibility, we gather data on the period before the LBO events and on matched private

firms during the life of the LBO and perform similar analyses.

5.1. Pre-LBO Behavior

To examine the LBO target's behavior prior to the deal, we search Dealscan for all deals for the same borrowers in our sample in the 10 years before their LBOs. This search yields 176 sample firms with prior loan data. We then hand-collect the ownership of these firms before the LBOs through Capital IQ and Pitchbook and exclude the 104 firms that had prior private equity ownership, leaving 72 firms that were not owned by private equity prior to the LBO and that have sufficient data available for the analysis. Summary statistics appear in Table 11, Panel A. Column (7) contains the

significance level of differences in means for pre- and post-LBO periods (excluding the LBO event) for firms where both are available.

This table shows that financing frequency was similar, but the probability of issuing a dividend is much higher after the LBO, at 1.86% of quarters during the LBO versus 0.7% of quarters prior. As in prior work, we find that the face value of the outstanding debt was markedly higher after the LBO, at 1,266 versus 676.4 prior to the LBO. Coupled with this, the proportion of revolving debt, *Revolver*, is much lower after the LBO, at 10.6% versus 50.2% prior to the LBO. Spreads and maturity are also significantly higher during the LBO. Changes at the time of refinancing are similar pre- and post-LBO, however, with the exception of the increase

Table 11. Pre-LBO Period

Panel A: Summary statistics							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Sample firms with pre-LBO period			Pre-LBO period			
Variables	Mean	p50	N	Mean	p50	N	(1)–(4)
<i>New_financing</i>	0.0951	0	1,451	0.108	0	1,691	
<i>Dividend</i>	0.0186	0	1,451	0.00710	0	1,691	***
<i>MnA</i>	0.0744	0	1,451	0.0378	0	1,691	***
<i>FacilityAmtOS</i>	1,266	1,035	138	676.4	500	183	***
<i>Early</i>	0.312	0	138	0.268	0	183	
<i>Revolver</i>	0.106	0.0783	138	0.502	0.504	183	***
<i>Spread</i>	4.335	4.250	109	2.456	2.156	137	***
<i>SpreadT</i>	4.285	4.274	135	2.947	2.704	138	***
<i>SpreadR</i>	3.630	3.750	115	2.317	2.136	179	***
<i>Maturity</i>	6.231	6.426	138	5.734	5.623	183	***
Δ <i>Spread</i>	0.0961	−0.00923	107	−0.00658	0	134	
Δ <i>SpreadT</i>	0.0288	0	135	−0.0850	0	129	
$\% \Delta$ <i>TotalDebt</i>	0.206	0.101	138	0.516	0.175	183	***
Panel B: Financing events and associated changes for pre-LBO period							
	(1)	(2)	(3)	(4)			
Variables	Financingevent	% Δ Debt	Δ Spreadall	Δ Spreadterm			
<i>SponsorFailure</i>	−0.550 (0.11)	−0.168 (0.82)	0.176 (0.76)	0.262 (0.58)			
<i>SponsorIPO</i>	−0.237 (0.58)	1.770* (0.08)	0.461 (0.17)	−2.400 (0.50)			
<i>SponsorNoHist</i>	−0.160 (0.21)	−0.0428 (0.79)	0.0689 (0.51)	−0.0797 (0.62)			
<i>Early</i>	−1.088*** (0.00)	0.621** (0.02)	−0.165 (0.22)	−0.209* (0.09)			
Observations	1,691	183	134	129			
R^2	0.288	0.639	0.862	0.829			
Controls	Yes	Yes	Yes	Yes			
FF49 fixed effects	Yes	Yes	Yes	Yes			
Year-quarter fixed effects	Yes	Yes	Yes	Yes			

Notes. In Panel A, The first three columns are summary statistics for the deal period for the 72 non-PE-to-PE deals for which Dealscan data are available for some span during the 10 years before the deal and leading up to the deal. The last three columns are data during the preperiod for these deals. Panel B presents principal regressions from the analysis for the period prior to LBOs. Some are omitted due to small samples. Column 1 presents all quarters, and the dependent variables are the incidence of a refinancing and the incidence of a refinancing that includes a dividend. Control variables not shown are *BankSponsor*, *PublicToPrivate*, *ClubDeal*, *lagRelationshipLenderC*, *lagRelationshipLenderS*, *lagRelatedLender*, *MnA*, *Dividend*, $\% \Delta$ *Totaldebt*, Δ *Maturity*, and *IndFailure*. Columns (2)–(4) are conditional on a financing event. Variable definitions appear in Table 2.

*, ** and ***Significance levels of 10%, 5%, and 1%, respectively.

in debt at the time of a financing event, which is higher pre-LBO.

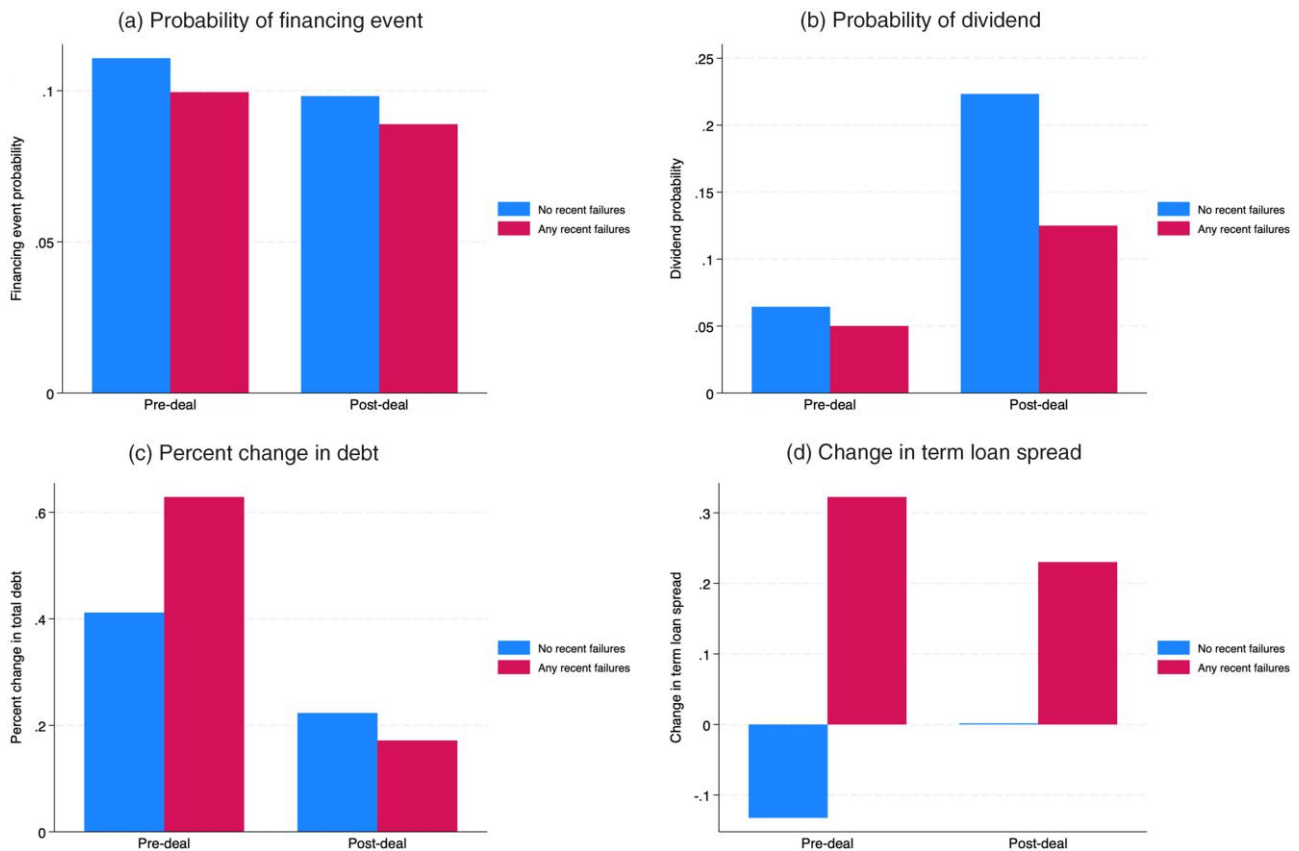
Figure 5 examines relations between pre- and post-LBO variables and levels of *SponsorFailure*. These further splits result in small sample sizes; for example, there are only 12 dividends in the pre-LBO period. This figure shows either no significant difference pre- and post-LBO for financing events in Figure 5(a) or an increase in dividend issuance for sponsors that do not have a recent failure after the LBO in Figure 5(b). In Figure 5(c), there is slightly less debt issued when sponsors have had a recent failure postdeal. In Figure 5(d), the relation between recent failures on the cost of the term loan is counterintuitively higher pre-LBO for sponsors that have had no failures, but this may be due to the larger increase in debt pre-LBO for sponsors with recent failures that appears in Figure 5(c). These results suggest that control variables are necessary. Table 11, Panel B, examines these relations statistically and replicates key regressions of the analysis using the pre-LBO data. As expected, we find no significant effect of *SponsorFailure* in the pre-LBO sample.

5.2. Matched Firms

We also construct a sample of comparable firms that are private but not owned by private equity. We start with deals with U.S. borrowers in Dealscan for which the sponsor and exchange ticker fields are blank, where the package currency is U.S. dollars, and where the deal purpose (including primary, secondary, and tertiary purpose) is not a leveraged buyout. To rule out financial borrowers, we further exclude borrowers in Standard Industrial Classification codes 6211, 6732, 6733, 6371, and 6798 and with borrower names containing “LP” or Roman numerals. With this group, we search for firms in the same Fama-French 49 industry as the sample firm and that have originated a similar-sized package in the same year and quarter as the sample firm. If there are several such firms, we choose the candidate that minimizes the difference in package size. We retain matches that greater than 50% smaller or less than 50% larger compared with the initial LBO package size, resulting in matches for 207 LBOs.

Summary statistics for these comparable firms for the LBO quarter appear in Panel A of Table 12. The first

Figure 5. (Color online) Pre- and Postdeal Averages of Key Variables for 72 LBOs



Notes. Sample is LBOs where predeal information is available leading up to the quarter prior to the buyout and where the target was not owned by private equity prior to the deal. Sample sizes are small. Panel (b) is based on 12 dividends during the pre-LBO period and 27 during the post-LBO period.

three columns present data from the LBO sample and the following three columns present data from the matching firms for all quarters of the LBO firms' sample periods. Column (7) presents significance levels of the paired *t*-tests between LBO firms and their matches at the LBO date. The results are consistent with the

analysis of the pre-LBO period. All measures of cost are significantly higher, with weighted average spreads averaging 4.312% versus 2.358% for matched firms and both revolvers (3.672% versus 2.012%) and term loans (4.456% versus 3.178%) being significantly more expensive. The proportion of revolving debt is significantly

Table 12. Matched Firms

Panel A: Summary statistics for the LBO quarter							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Sample firms with matches			Matched firms			
Variables	Mean	p50	N	Mean	p50	N	(1)–(4)
FacilityAmtOS	514.7	300	207	417.6	264.8	207	**
Spread	4.312	4.250	177	2.358	2	168	***
SpreadT	4.456	4.435	185	3.178	2.720	92	***
SpreadR	3.672	3.500	179	2.012	1.750	142	***
Revolver	0.191	0.135	207	0.546	0.541	207	***
Maturity	6.568	7	207	4.711	5	207	***
MaturityT	6.337	6.750	195	5.106	5	111	***
MaturityR	5.162	5	200	4.360	5	163	***
Panel B: Summary statistics for post-LBO quarters							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Variables	Mean	p50	N	Mean	p50	N	(1)–(4)
New_financing	0.0942	0	3,024	0.0513	0	3,024	***
Dividend	0.0109	0	3,024	0.000992	0	3,024	***
MnA	0.0479	0	3,024	0.0721	0	3,024	
Early	0.316	0	285	0.342	0	155	
%ΔTotalDebt	0.211	0.111	284	0.743	0.415	155	***
ΔSpread	−0.151	−0.0783	235	0.0331	0	117	**
ΔSpreadR	−0.0634	0	242	0.0386	0	117	
ΔSpreadT	−0.168	−0.0322	261	−0.0809	0	61	
ΔRevolver	−0.00772	−0.00484	284	0.00767	0	151	***
ΔMaturity	−0.325	−0.279	284	0.0656	0	155	***
ΔMaturityT	−0.487	−0.500	263	−0.323	0	70	
ΔMaturityR	−0.462	−0.333	268	−0.176	0	136	***
Panel C: Changes conditional on financing events							
	(1)	(2)	(3)	(4)			
Variables	Financing event	%ΔDebt	ΔSpread all	ΔSpread revolver			
SponsorFailure	−0.630** (0.03)	−0.594 (0.74)	0.518 (0.55)	0.0631 (0.89)			
SponsorIPO	0.0835 (0.82)	−3.435 (0.16)	−1.558* (0.06)	−2.034*** (0.01)			
SponsorNoHist	−0.0324 (0.80)	−0.325 (0.22)	−0.508** (0.05)	−0.195 (0.48)			
Early	−1.664*** (0.00)	1.584*** (0.00)	0.337** (0.04)	0.324*** (0.01)			
Observations	3,024	155	117	117			
R ²	0.399	0.661	0.755	0.871			
Controls	Yes	Yes	Yes	Yes			
FF49 fixed effects	Yes	Yes	Yes	Yes			
Year-quarter fixed effects	Yes	Yes	Yes	Yes			

Notes. The 207 matched firms are private, non-private-equity-owned firms in the same Fama-French 49 industry with financing events of similar size in the same quarter as the LBO. In Panels A and B, the first three columns present data from the LBO sample and the following three columns present data from the matching firms for all quarters of the LBO firms' sample periods. Column (7) presents paired *t*-tests for Panel A and unpaired *t*-tests for Panel B. Panel C presents placebo regressions with matched firms. Control variables not shown are *BankSponsor*, *PublicToPrivate*, *ClubDeal*, *lagRelationshipLenderC*, *lagRelationshipLenderS*, *lagRelatedLender*, *MnA*, *Dividend*, *%ΔTotaldebt*, *ΔMaturity*, and *IndFailure*.

*, ** and ***Significance levels of 10%, 5%, and 1%, respectively.

Table 13. Do Recent Sponsor Failures and Changes in Terms at Financing Events Predict LBO Failures?

Variables	(1) Failure Inception	(2) Failure Financing event	(3) Failure Financing event	(4) Failure Financing event	(5) Failure Financing event
<i>SponsorFailure</i>	−0.393 (0.78)	1.468** (0.01)	1.301** (0.04)	0.473 (0.60)	0.400 (0.65)
<i>SponsorNoHist</i>	−0.633 (0.19)	−0.927** (0.03)	−1.019** (0.03)	−1.205** (0.02)	−1.329** (0.01)
$\Delta Spread$			0.382** (0.02)		
$\Delta SpreadT$				0.387** (0.01)	0.458*** (0.00)
$\% \Delta TotalDebt$					−1.690*** (0.00)
$\Delta Maturity$					0.208 (0.21)
<i>SponsorIPO</i>	2.825 (0.30)	2.633** (0.03)	2.630* (0.06)	1.975 (0.23)	2.115 (0.20)
Observations	411	663	512	613	613
R^2	0.658	0.526	0.551	0.598	0.623
Controls	Yes	Yes	Yes	Yes	Yes
FF49 and year-quarter fixed effects	Yes	Yes	Yes	Yes	Yes
Linear coefficient <i>SponsorFailure</i>	−0.023	0.176	0.132	0.041	0.034
Linear p <i>SponsorFailure</i>	(0.90)	(0.11)	(0.32)	(0.62)	(0.67)

Notes. Probit model. Columns (1) and (2) present inception for those deals that have exited as of the end of our sample period in 2022. Columns (3)–(8) present the same deals, but the prediction is at the time of intermediate financing events. Control variables are *Opaque*, *ClubDeal*, *BankSponsor*, *RelationshipLenderS*, *RelationshipLenderC*, *RelatedLender*, *PublicToPrivate*, *Debt_EV*, *LogEV* and *IndFailure*. Standard errors are clustered by time in quarters for the Probit model and by time in quarters and by Fama-French 49 industry classification for the linear model.

*, ** and ***Significance levels of 10%, 5%, and 1%, respectively.

lower at 0.191 versus 0.546 for matched firms, and measures of maturity (6.568 versus 4.711 years for weighted average maturity) are significantly higher for LBO firms than for matched firms.

Panel B proceeds to examine changes in subsequent financings. We use unpaired *t*-tests because each LBO firm and its match have subsequent financing events at different times and frequencies. This panel shows that new financing events and dividends occur more often in LBO firms compared with matched firms (9.42% of quarters versus 5.13% of quarters for financings). The proportion increase in debt at the time of a financing event is higher in matched firms. As one might expect, because LBO sponsors are looking to exit, maturity decreases more in post-LBO financings compared with matched firm financings. Perhaps because of the small sample and the unpaired tests, we do not find any other significant differences at the time of refinancing.

Panel C presents a probit model for a financing events and ordinary least squares (OLS) models for the change in financing costs. Although there is a negative relation with the incidence of financing events, the relation between *SponsorFailure* and refinancing costs is not present in the matched sample. One caveat is that, because our LBO firms refinance at twice the rate of the matched firms and use a greater proportion of term loans, there are more observations in the LBO firm regressions. In

this analysis, we only show regressions where the number of observations exceeds the number of regressors.

5.3. Other Robustness Tests

Other robustness tests appear in the Online Appendix. Our measure of *SponsorFailure* is measured in the last two years or since the last financing event. Table B1 examines other measures of sponsor failures in the last one, two, and three years, respectively. Table B2 removes all LBOs that end in failure. Table B3 restricts the sample to one financing per LBO. In all three cases, the results are similar.

6. Does Deterioration in Financing Terms Predict Deal Failure?

This section examines whether lenders correctly increase lending rates after recent sponsor failures. As a back-of-the-envelope calculation, we saw in Table 10 that a one-standard-deviation increase in *SponsorFailure* is associated with an 8.5-basis-point increase in spread, which amounts to 50 basis points of increased revenue to the lender over the following six years, the average life of a loan. According to column (1) of Table 4, a one-standard-deviation increase in sponsor failure over the last eight quarters is associated with a $0.0772 \times 0.181 = 0.0125$ increase in sponsor failure over the following eight

Table 14. Picking vs. Managing?: Competitiveness of Initial LBO Bidding Conditions

Variables	(1) ΔDebt	(2) ΔCost	(3) ΔSpread	(4) ΔSpread term
<i>SponsorFailure</i> × <i>Competitive</i>	−0.172** (0.04)	0.343 (0.24)	0.324 (0.16)	0.496*** (0.01)
<i>SponsorFailure</i> × <i>NotCompetitive</i>	−0.144** (0.04)	0.410 (0.13)	0.376* (0.09)	0.568*** (0.00)
<i>SponsorIPO</i>	0.0706 (0.46)	0.210 (0.53)	0.104 (0.73)	0.226 (0.43)
<i>SponsorNoHist</i>	−0.0534** (0.03)	−0.0255 (0.70)	−0.0949* (0.06)	−0.0188 (0.79)
Observations	803	564	593	748
R ²	0.423	0.352	0.329	0.275
Controls	Yes	Yes	Yes	Yes
FF49 and year-quarter fixed effects	Yes	Yes	Yes	Yes

Notes. Competitive initial LBO conditions are defined as the net proportion of banks tightening being less than equal to zero. Slow periods are defined as times when the net proportion is positive. Control variables are *Opaque*, *ClubDeal*, *BankSponsor*, *RelationshipLenderS*, *RelationshipLenderC*, *RelatedLender*, *PublicToPrivate*, *Debt_EV*, *LogEV* and *IndFailure*. Standard errors are clustered by time in quarters for the Probit model and by time in quarters and by Fama-French 49 industry classification for the linear model.

*, ** and ***Significance levels of 10%, 5%, and 1%, respectively.

quarters. If we apply this probability to the lender’s term loan, a loss-given-failure of $0.0051/0.014 = 36.4\%$ would reconcile the increased revenue to the expected incremental loss (for a risk-neutral lender). Hotchkiss et al. (2021) find in their 1997–2010 sample that private equity (PE)-backed firms resolve financial distress more quickly and are more likely to retain control than non-PE-backed firms, which may result in a reasonably high recovery rate for lenders. Because the failures in our study include restructurings and recapitalizations, as well as bankruptcies, this estimated loss-given default is plausible.

Table 13 examines this question statistically, using our sample of the 411 exited deals as of the end of our sample period. Predictably, given the result in Table 4 that *SponsorFailure* is not autocorrelated beyond two years, column (1) shows that the level of *SponsorFailure* at LBO inception does not predict whether the deal ultimately fails. Column (2) shows that, at the time of an intermediate financing event, however, *SponsorFailure* is predictive of whether the deal fails. The remaining columns examine whether lenders correctly adjust for this increased risk and find that they do. In columns (3)–(5), controlling for the changes in lending terms at the time of the refinancing events subsumes the effect of *SponsorFailure* on the ultimate failure probability of the deal. This suggests that lenders do accurately adjust for changes in the sponsor’s probability of borrower default by adjusting loan terms at the refinancing time.

7. Do Lenders Focus More on Sponsors’ Target Picking or Managerial Abilities?

One may wonder whether the effect on lending decisions is related to the sponsor’s time-varying ability to pick LBO targets or to their time-varying skill at managing companies and implementing operational

improvements. To shed light on this question, we use the finding by Kaplan and Schoar (2005) that fundraising is especially high for LBOs when the difference between the earnings yield on the S&P 500 and the high yield rate is large. During these times, purchasing a company and financing it with debt is especially attractive. With many sponsors competing for a target and bidding up the price, it is likely that the sponsor with the highest value placed on operational improvements will win the deal. If lenders believe that sponsors’ managerial ability is of greatest interest, one might expect the effects of *Sponsorfailure* to be concentrated in *Competitive* times.²² Table 14 presents the main results in *Competitive* versus *NonCompetitive* periods. We find that the results are of similar magnitude and significance for both periods, suggesting that a combination of sponsors’ picking and managerial ability is likely to drive the result or that these measures are very correlated among sponsors.

8. Conclusion

In this paper, we uncover new facts about the intermediate financing behavior of LBOs. We reconstruct the time series of the debt structure of 510 U.S. LBO targets and find that almost half of the subsequent financing deals present a clear loosening of lending terms, whereas only 6.8% present a tightening of terms. Moreover, 67% of financing activity occurs in the quarter prior to the maturity of the earliest maturing loan. The occurrence and timing of these deals is plausibly determined by the upcoming maturity of the debt and not by distress.

Next, we test whether recent publicly available signals of sponsor skill at avoiding poor outcomes affect financing terms. If lenders focus on sponsors’ reputation for not expropriating lenders (Malenko and

Malenko 2015) or downside risk taken by the sponsor, they may penalize recent failures in the sponsors' portfolio. Alternatively, they may prefer to favor banking relationships in order to drive other business with the bank or not focus on these measures at all. Our measure of recent failures is the proportion of the sponsor's exited deals over the last two years that have resulted in bankruptcies, recapitalizations, and restructurings, which are usually adverse outcomes for lenders. This variable displays short-term autocorrelation within sponsor over nonoverlapping two-year periods, but further lags are not significantly related, justifying a lender's focus on recent deal outcomes.

Our time series approach allows us to control for unobserved variables that are fixed within an LBO deal and a host of time-varying controls and fixed effects. We examine the incidence of new financing activity within LBO deals and find that recent sponsor failures are associated with a lower probability of improvement in financing terms. When we examine loan amount, maturity, and cost in turn, we find that effects are driven primarily by loan cost. A one-standard-deviation increase in our measure of recent sponsor failures is associated with an 8.5-basis-point increase in total loan spread. This increase is primarily driven by term loans, where the effect is 12.3 basis points.

Studying LBOs offers a unique opportunity to dissociate the effects of management, which the LBO sponsor provides, from the characteristics of the underlying target in determining capital structure dynamics throughout the LBO holding period. More broadly, our study sheds light on the relative importance of perceptions of managerial skill compared with cooperative behavior induced by relationships and the underlying financial situation of borrowing firms.

These findings have several implications. Economic models of LBOs should consider that intermediate financing activity adds nonnegligible value for equity holders. Further, in assessing optimal leverage levels for a given LBO and its response to a distressed situation, a sponsor should consider the effects of potential default on the near-term financing opportunities of other companies across their funds' portfolios. These financing effects may decrease the diversification benefits of holding many portfolio companies contemporaneously. More broadly, our study sheds light on the relative importance of perceptions of managerial skill compared with cooperative behavior induced by relationships and the underlying financial situation of borrowing firms.

Acknowledgments

The authors thank Bo Becker, David Feldman, Paul Schultz, two anonymous referees, seminar participants at the University of Notre Dame, the University of Utah,

and Georgia State University, and participants of the 2021 Private Equity Research Consortium (PERC) symposium at the University of North Carolina, Chapel Hill for helpful comments. The authors also thank Justin Braun and Michael Kusznerko for excellent research assistance.

Endnotes

¹ In their review of the literature, Colla et al. (2020, p. 211) state: "As of the time of this survey, our understanding of the debt structure of private firms is quite limited, despite the fact that private firms form a substantial part of the global economy." See also Roberts and Sufi (2009), Denis and Wang (2014), and Badertscher et al. (2019). Dangl and Zechner (2021) discuss refinancing costs and roll-over risks (Admati et al. 2018, Demarzo and He 2021).

² "TPG's Aleris in bankruptcy protection," *Financial Times*, March 12, 2009.

³ Indap and Frumes (2021).

⁴ "iHeartMedia wins legal fight," *Wall Street Journal*, May 24, 2016.

⁵ "Asset transfers leave creditors feeling 'J Screwed'" *Financial Times*, June 5, 2020.

⁶ "Why a recent legal decision may upend private equity," *New York Times*, February 28, 2021.

⁷ "Loan documents deteriorate further following Neiman Marcus lawsuit," *Reuters*, April 24, 2019.

⁸ "PetSmart moves part of Chewy.com out of creditors' reach," *Bloomberg*, June 4, 2018.

⁹ Cohn et al. (2014, 2022) find no reduction of leverage after the LBO in confidential tax records for samples ending in 2007 and 2009. Ivanov et al. (2021) find that leverage increases following tax cuts for both privately held and publicly listed firms, but very few of their firms are private equity owned.

¹⁰ Drake et al. (2014) find that between 2008 and 2013, Cerberus and Apollo had the highest portfolio default rates at 15.9% and 12.9%, whereas Madison Dearborn had the lowest at 1.5%, which does not suggest a strong inverse relation between default rates and LBO assets under management.

¹¹ Results are robust to looking over one or three years, as shown in the Online Appendix, Table B1.

¹² Almeida et al. (2011) and Choi et al. (2018) use maturing debt as an instrument for refinancing for publicly traded firms.

¹³ Boot (2000) provides an early review of this vast literature. For example, Bharath et al. (2007) find that the probability of a relationship lender providing a future loan is significantly higher than that of a nonrelationship lender, and Bharath et al. (2011) and Zhang et al. (2023) find that relationship lending results in better loan terms. Chu (2018) finds that increasing conflict between equity holders and creditors can lead to higher dividend payouts, in particular, for distressed companies.

¹⁴ See Section 2.2.2.

¹⁵ For example, in the 2007 acquisition of Pinnacle Foods, the new holding company sponsored by Blackstone Capital Partners V assumed 860M of Pinnacle's debt, and there is no evidence that this was immediately refinanced.

¹⁶ For example, Tegrant Corporation in Preqin is Svenska Cellulosa AB-North Ops in SDC. Both were acquired for \$400M by Metalmark Capital on January 23, 2007. A news search confirms that these are one and the same. The name was changed to Tegrant on March 27, 2007.

¹⁷ For example, Sungard Data Systems was taken private in 2005 by a consortium that included Goldman Sachs. The debt was provided

by a syndicate that also included Goldman Sachs. Pinnacle Foods was purchased by J.P. Morgan Partners and Prospect Hill Growth Partners. According to Dealscan, the debt was provided by a syndicate that included J.P. Morgan. We consider both these deals to be parent-financed bank deals, which are both related lender deals and bank-financed deals. An example of a related lender deal that is not bank financed is TeamHealth, purchased in a club deal led by Blackstone in 2016. The debt providers include Blackstone, according to Dealscan.

¹⁸ A “bank” here is not necessarily under the umbrella of a bank holding company. For example, Goldman Sachs and its affiliates are considered a bank here throughout our sample period and not just since 2008. This definition contrasts with the narrower definition of banks that is used in the growing literature on nonbank lending. In that literature, nonbank lenders often include nondepository lending arms of investment banks.

¹⁹ For example, Irving Place Capital was formerly known as Bear Stearns Merchant Banking, and EagleTree Capital is the current name (Preqin uses current names) of a private equity firm that was spun off from Wasserstein Perella in 2001, so we consider these to be bank sponsors. One Equity Partners was part of J.P. Morgan until 2015, and J.P. Morgan still retains some ownership.

²⁰ One caveat is that we remove deals that have no loans at inception in case the loans are missing from Dealscan, so we would miss a deal that is exclusively financed with bonds and perhaps a bank loan that is not syndicated.

²¹ The *margins* command in Stata.

²² Earnings yield data come from Aswath Damodaran’s website, and the high-yield rate is from the St. Louis Federal Reserve’s website: data item code *BAMLH0A0HYM2EY*.

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